Assessing Bermudan exercise policies: numerical error and confidence coverage

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20 October 2026

Abstract

I examine whether a fitted Bermudan exercise policy meets a monetary tolerance by separating policy shortfall, martingale slack, sampling uncertainty and numerical integration error. In a directly integrable two-right model, 5,000 repetitions with 2,048 paths give 96.50 per cent coverage for nominal 97.5 per cent Student upper limits, despite zero dual slack. Clipping with an analytical tail allowance raises observed coverage to 99.98 per cent and reduces mean excess by 85.3 per cent relative to Cantelli. Across eight five-asset fits, gain-directed refinement meets a realised gap-width target of 0.01 at a median 75.7 per cent of the matched common-tolerance implementation's reconstructed CPU budget. A separate evaluation with 16,384 costed paths per fit produces upper limits below one unit for every policy; half-unit targets remain unresolved. The monetary interpretation requires the stated sampling and ideal-observation enclosure premises, beyond the supplied-binary-state calculations certified here.

Keywords: Bermudan options; martingale duality; confidence bounds; interval arithmetic; Monte Carlo.
JEL classification: G13; C15; C63.

1 The assessment problem

A pricing team has fitted an exercise rule and needs to decide whether further training is worthwhile. A simulation upper bound exceeds the estimated policy reward by a small amount. Is that amount a reliable upper limit on what the rule sacrifices? Three quantities enter the answer: the actual shortfall relative to optimal exercise, the slack introduced by the evaluation martingale, and the uncertainty in estimating their sum. Numerical error in conditional expectations introduces a fourth quantity. Reducing any one of them need not reduce the others.

I study when these distinctions affect a usable assessment. The one-dimensional problem serves as a validation instrument: it supplies a numerical reference and, with two rights, a directly integrable shortfall. The multidimensional problem supplies the setting in which simulation becomes useful. The intended decision is whether an independent upper limit on ℓ_π lies below a declared monetary tolerance η, at what total computational cost, and whether a cheaper evaluator reaches the same conclusion. I examine verified conditional integration and refinement driven by the stopped gain, rather than treating every conditional expectation as requiring the same accuracy.

A useful assessment must also reach a decision at a stated monetary tolerance. I therefore distinguish coverage of an expectation from the ability to certify an acceptable policy. The two-right comparison includes direct verified quadrature, a root-free analytical bound and three simulation assessors. These deterministic alternatives prevent a favourable simulation comparison from excluding the simplest usable calculation.

I contribute a comparative evaluation design for deciding when a tight numerical assessment supports a useful monetary conclusion. The two-right diagnostic separates sampling failure

from dual slack and tests a tail-controlled response against a known shortfall. The multiasset experiments compare numerical work at a common realised gain-width target, retaining cheaper moment-based competitors and their looser gaps. The monetary experiment then compares upper limits with declared tolerances for the same frozen policies. These comparisons address reliability, numerical cost and decision usefulness separately: savings in conditional integration do not by themselves establish cheaper policy certification.

The foundations are established. Rogers (2002) and Haugh and Kogan (2004) develop martingale dual valuation; Andersen and Broadie (2004) construct implementable primal-dual simulation for multidimensional contracts. I use Longstaff and Schwartz (2001) as the exercise-policy baseline. Regression-later centring and tractable conditional basis moments are central to Glasserman and Yu (2004); Belomestny et al. (2009) develop true non-nested upper bounds and use max-call information in their numerical bases. I use these established constructions as comparators. The max-call benchmark also appears in Becker et al. (2019), whose experiments extend to 500 assets; the present five-asset results do not establish comparable scalability.

The more specific issue is computable numerical error in a stopped gain. Brown et al. (2010) analyse information-relaxation penalties and their perturbation, while Haugh and Kogan already adapt deterministic conditional calculations using successive estimates. I replace that numerical stopping criterion with an actual enclosure width and retain cancellation between shared increments before allocating refinement. Stopping-time refinements and continuation-region calculations also appear in Joshi and Tang (2014) and Ibáñez and Velasco (2020). Establishing a useful distinction requires the measured comparison in Section 6. The general duality and concentration inequalities used here retain their original attribution.

I collect the notation in Table 1, separating the financial target, the evaluation martingale and the numerical and sampling controls.

Table 1: Notation and its role in the assessment

Symbol	Meaning
d, J, t_j	Number of assets, last exercise index and calendar dates.
$\mathcal{D}, \mathcal{F}_j$	Frozen fitting information and information available at date t_j.
H_j, τ, V_π, V^*	Discounted reward, policy stopping index, policy value and optimal value.
M_j, U, G_π, s_M	True martingale, upper payoff, stopped gap and expected dual slack.
P_j, F_j	Conditional European value and discounted reward advantage $F_j = H_j - P_j$.
$v_j, c_j, \widehat{c}_j$	Frozen fitted value, its exact conditional expectation and recorded numerical centre.
$[a_j, b_j]$	Enclosure of the conditional centring error $c_j - \widehat{c}_j$.
n, R, α	Evaluation paths, independent coverage repetitions and one-sided error probability.
$B, T_B, \eta, \varepsilon$	Clipping cap, omitted-mean allowance, monetary shortfall tolerance and numerical gain-width target.

2 Model, policy and target

Under a maintained pricing measure $\mathbb{Q}$, the asset vector follows

$$dS_t^i = (r - q_i)S_t^i\,dt + \sigma_i S_t^i\,dW_t^i, \qquad d\langle W^i, W^k\rangle_t = \rho_{ik}\,dt. \tag{1}$$

For a Bermudan max-call, the discounted reward at an exercise right is

$$H_j = e^{-rt_j}\left(\max_{1\le i\le d} S_{t_j}^i - K\right)^+. \tag{2}$$

The calendar is finite. I write $S_j = S_{t_j}$ and $\Delta t_j = t_j - t_{j-1}$. The class $\mathcal{T}$ contains the stopping indices taking values in $\{0, \dots, J\}$ for the filtration $(\mathcal{F}_j)$. A policy selects $\tau \in \mathcal{T}$ using only

information available at each right. Its value is $V_\pi = \mathbb{E}[H_\tau \mid \mathcal{D}]$, and

$$V^* = \sup_{\nu \in \mathcal{T}} \mathbb{E}[H_\nu \mid \mathcal{D}], \qquad \ell_\pi = V^* - V_\pi. \tag{3}$$

I condition on fitted coefficients, basis choices and tie rules. Evaluation paths are independent of $\mathcal{D}$. The filtration includes that frozen information at its initial date, which makes the fitted policy adapted on the independent evaluation sample. This conditioning matters: the present assessment concerns a particular fitted policy, rather than the average result of a random training procedure. The continuation fit estimates a conditional mean. Appendix H constructs that expectation and explains the pricing-measure convention; Appendix I connects it to the regression and PDE implementations.

I fit the baseline policy by backward least-squares continuation regression. Its basis contains all monomials through degree two in the descending sorted coordinates $S^i/K - 1$. Every coordinate enters, and exchangeability makes asset permutation immaterial. At a nonterminal right, the policy exercises only when the discounted reward strictly exceeds the continuation estimate clipped below at zero. Ties continue; survivors exercise at expiry. Insufficient in-the-money observations or deficient regression rank produce continuation instead of an unstable fitted decision. [1]

The benchmark process uses $r = 0.05$, $q_i = 0.10$, $\sigma_i = 0.20$, $K = 100$, $T = 3$ and independent Brownian motions, with initial states 90, 100 and 110. Its ten rights are $t_j = jT/9$, $j = 0, \ldots, 9$. These are published mathematical benchmark parameters, not observed option quotes. I simulate exact GBM transitions at the rights, so there is no Euler time-step bias in the ideal transition law. Floating-point simulation and regression remain numerical computations.

3 What the primal-dual gap measures

Let $M_0 = 0$ be an integrable martingale, conditional on the frozen fit. Define

$$U = \max_{0 \le j \le J} (H_j - M_j), \qquad G_\pi = U - H_\tau + M_\tau \ge 0. \tag{4}$$

Finite-horizon optional sampling gives $\mathbb{E}[M_\tau \mid \mathcal{D}] = 0$. Therefore

$$\mathbb{E}[G_\pi \mid \mathcal{D}] = \ell_\pi + s_M, \qquad s_M = \mathbb{E}[U \mid \mathcal{D}] - V^* \ge 0. \tag{5}$$

This is the established dual interpretation. A tighter martingale can decrease s_M without changing a single exercise decision. Conversely, a better policy can decrease ℓ_π under a common upper payoff. I report these changes separately whenever an independent reference permits it.

A numerical reference must solve the same calendar and financial problem. Its grid resolution and truncation domain are checked separately. The initial price returned by an approximate value grid, the reward of the forward policy induced by that grid, and the upper estimate constructed from it are distinct objects. Their convergence rates need not agree. The one-dimensional deterministic grid provides an inspectable reference. Appendix I separates discretisation, boundary and interpolation errors.

3.1 A diagnostic with an explicit loss target

I retain the one-asset process with $S_0 = 100$ but change the calendar to $t_1 = 1.5$ and $t_2 = T = 3$. The policy always waits until T. The diagnostic uses a reduced exercise calendar.[2] Write

$$C_1 = \mathbb{E}[H_2 \mid \mathcal{F}_1], \qquad V_E = \mathbb{E}[H_2]. \tag{6}$$

[1] This policy basis differs from that used by Andersen and Broadie (2004).

[2] The published benchmark has ten exercise rights; this diagnostic has two.

The exact two-right value is $\mathbb{E}[\max(H_1, C_1)]$. The policy shortfall consequently equals $\mathbb{E}[(H_1 - C_1)^+]$. Choosing $M_1 = C_1 - V_E$ and $M_2 = H_2 - V_E$ makes the stopped gap

$$G_{\text{centred}} = (H_1 - C_1)^+, \tag{7}$$

with zero theoretical dual slack. This provides a target against which coverage can be measured independently of a PDE approximation. With $M = 0$, the gap is $(H_1 - H_2)^+$ and includes positive dual slack.

I calculate both expectations by one-dimensional lognormal quadrature. Appendix A gives the conditioning identities and numerical error treatment. These calculations supply point references; the threshold-policy references below use interval-certified integration.

The expiry value is 6.020789 and the two-right optimum is 7.177780. Thus $\ell_\pi = 1.156991$. The zero-martingale expected gap is 2.790257, with slack 1.633265. This difference is large enough to conceal an inference failure if coverage of the smaller loss is substituted for coverage of the gap's own expectation.

4 Sampling reliability and a finite-sample response

For each sample size $n \in \{8, 32, 128, 512, 2048\}$, I draw 5,000 independent repetitions. Within each repetition, the two martingale gaps use matched paths. The design and random streams are retained before execution. Coverage is the fraction of upper limits exceeding the appropriate expectation target; Wilson intervals describe uncertainty in that observed fraction. The limits are pointwise, without a simultaneous claim across methods or sample sizes.

The ordinary one-sided Student limit is $\bar{G} + t_{n-1,0.975} s_G/\sqrt{n}$. Its use relies on approximate sampling calibration. In this diagnostic the gap has a substantial atom at zero and a right tail. At small samples, zero-only repetitions supply a zero estimated sampling radius despite a positive expectation. The observed coverage, rather than the shape description alone, establishes the failure. Table 2 reports the frequencies at each sample size; Figure 1 places their Wilson intervals against the nominal target.

Table 2: Observed coverage of the centred-gap upper limits (per cent)

Paths	Student	Clipped EB	Clipped Hoeffding	Cantelli
8	72.48	100.00	100.00	100.00
32	89.72	100.00	100.00	100.00
128	94.88	100.00	100.00	100.00
512	95.78	100.00	100.00	100.00
2,048	96.50	99.98	100.00	100.00

Each entry uses 5,000 independent repetitions and targets the two-right shortfall 1.156991. Nominal one-sided coverage is 97.5 per cent. The clipping cap is frozen using an independent pilot.

At $n = 2048$, centred Student coverage is 96.50 per cent; its 95 per cent Wilson interval is $[95.954, 96.975]$ per cent. That interval lies below nominal coverage. The zero-martingale gap can nevertheless cover the policy loss much more frequently because its own expectation exceeds the loss. The resulting apparent protection is dual slack, rather than accurate inference.

4.1 Clipping with an independently selected cap

For $B \geq 0$, define $Y = \min(G, B)$. Suppose a deterministic allowance satisfies $T_B \geq \mathbb{E}[(G-B)^+]$. Then $\mathbb{E}[G] \leq \mathbb{E}[Y] + T_B$. A bounded-observation inequality for $\mathbb{E}[Y]$ therefore extends to an upper limit for the unbounded gap mean.

For the centred diagnostic, conditional Jensen gives a useful shifted envelope. Set $h = T - t_1$ and

$$a = e^{-rt_1}(1 - e^{-qh}), \qquad c = e^{-rt_1} K (1 - e^{-rh}). \tag{8}$$

For $r, q \geq 0$, the conditional call lower bound implies

$$G_{\text{centred}} \leq (aS_1 - c)^+, \qquad T_B = \mathbb{E}[(aS_1 - c - B)^+]. \tag{9}$$

I obtain this envelope by comparing the immediate call payoff with the conditional Jensen lower bound for continuation, splitting according to the sign of that lower bound. I use it because the subtraction of c reduces the omitted-mean allowance compared with aS_1. The right-hand expectation is a scaled lognormal call and is calculable without fitting a tail distribution. The zero-martingale envelope is its immediate payoff $e^{-rt_1}(S_1 - K)^+$.

For $B > 0$, applying the empirical Bernstein inequality of Maurer and Pontil (2009, Theorem 4) to Y/B, with unbiased sample variance s_Y^2, gives

$$\mathbb{E}[G] \leq \bar{Y} + \sqrt{\frac{2s_Y^2 \log(2/\alpha)}{n}} + \frac{7B \log(2/\alpha)}{3(n-1)} + T_B \tag{10}$$

with probability at least $1 - \alpha$ under ideal iid sampling. I also compare Hoeffding's bounded-observation limit and Cantelli's limit based on a known second-moment bound. Their roles are different: Hoeffding uses the range, empirical Bernstein uses observed bounded variance, and Cantelli uses an analytical raw second-moment bound.

An independent 8,192-path pilot selects the cap from a fixed grid by minimising its forecast upper limit. I freeze the cap before evaluation. Conditional on the pilot, the concentration inequality still applies to fresh observations; poor cap selection can increase width but does not invalidate that mathematical argument. Evaluation replays the retained diagnostic streams, independently of the pilot. Pilot and cap-selection costs are included in the incremental comparison; the earlier oracle calculation is not charged again. The degenerate cap $B = 0$ is handled directly through its deterministic tail allowance.

At $n = 2048$, the empirical Bernstein cap for the centred gap is 12, with tail allowance 0.0500554. Its mean upper limit is 1.407599, compared with the loss 1.156991. It covers in 4,999 of 5,000 repetitions. Cantelli's mean upper limit is 2.859916. The mean excess above the target falls from 1.702925 to 0.250608, a reduction of 85.3 per cent. The clipped bound is conservative in this test, but substantially more informative. The mathematical coverage argument concerns the stipulated process and frozen pilot; floating-point evaluation of the tail formula is a separate numerical issue. Figure 2 compares conservatism across path counts. Its narrower Student limits must be read alongside the coverage failures in Figure 1.

4.2 From a mean bound to a monetary decision

I fix five policies before assessment. Four exercise at the first right only when $S_1 > \theta$, for $\theta \in \{100, 110, 120, 130\}$; the fifth always waits. The decision is to certify $\ell_\theta \leq \eta$, for $\eta \in \{0.10, 0.25, 0.50\}$ in discounted monetary units per share. A failed certification leaves the assessment unresolved. It does not establish that the policy is unacceptable.

Write $D(s) = H_1(s) - C_1(s)$. For the stated positive-dividend model, D is strictly increasing above the strike and has a unique exercise boundary $b > K$. Appendix A.2 derives this property. The exact terminal-call martingale gives

$$G_\theta = D(S_1)^+ \mathbf{1}_{\{S_1 \leq \theta\}} + (-D(S_1))^+ \mathbf{1}_{\{S_1 > \theta\}}, \qquad \mathbb{E}[G_\theta] = \ell_\theta. \tag{11}$$

A threshold policy disagrees with optimal exercise only between θ and b. Monotonicity of D on that interval therefore gives $G_\theta \leq |D(\theta)|$ for finite $\theta \geq K$. I use this elementary model calculation before simulation: at $\theta = 110$, the verified upper bound is 0.11050443, which already certifies both larger tolerances. The bound requires neither the boundary nor quadrature.

I separately enclose each population shortfall by verified integration over the interval where the threshold and optimal decisions disagree. Appendix A.2 gives the boundary enclosure and

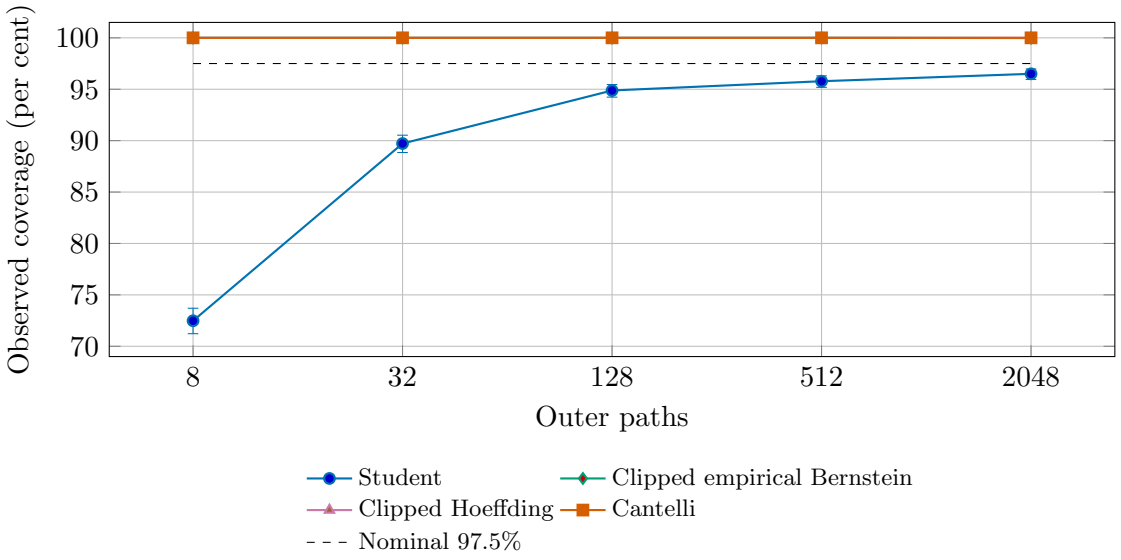


Figure 1: Coverage for the exactly centred terminal-call martingale and the always-wait policy, with 5,000 independent repetitions per path count. Bars are Wilson 95% intervals. Clipping caps are selected on an independent pilot; coverage is measured against the disclosed numerical policy-loss target.

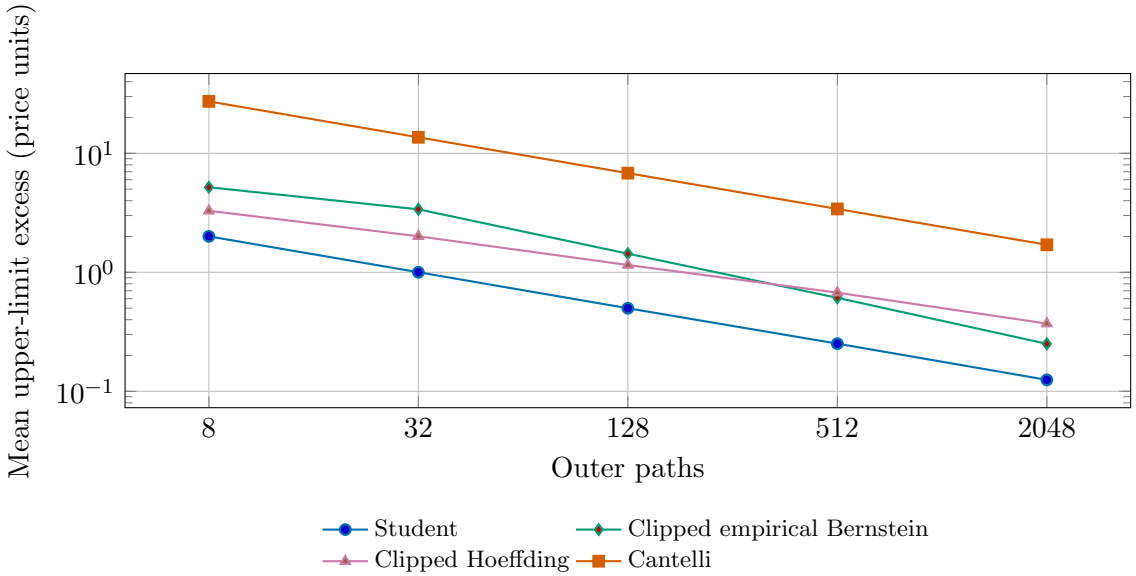


Figure 2: Mean upper-limit excess above the two-right policy-loss target 1.156991, under the same simulation design. Both axes are logarithmic. The ordinate measures excess above the target, rather than an interval width.

the allowance for its uncertain strip. This direct reference resolves every policy/tolerance pair, including rejection of all three tolerances for always waiting.

The simulation comparison uses zero, stock and terminal-call martingales. For the stock construction, $M_j = k\{e^{-(r-q)t_j}S_j - S_0\}$ with $0 \le k \le e^{-qT}$. An independent 8,192-path pilot freezes the coefficient by minimising its mean upper payoff and freezes clipping caps for the subsequent fixed looks. It uses neither the optimal reference value nor assessment outcomes. Analytical lognormal envelopes supply the omitted-mean allowances. Finite-threshold terminal-call gains instead use the verified support bound and require no pilot or terminal-state simulation.

I assess each frozen construction in 256 independent repetitions, with sample sizes of 128,

512, 2,048, 8,192 and 32,768. First- and second-period innovations have separate streams; assessors and policies share their matched innovations, and the looks use nested prefixes. Applying Equation (10) with a probability allocation no greater than $0.025/(5 \cdot 3 \cdot 5)$ protects the complete family under ideal independent sampling. The three tolerances reuse each upper limit. The first successful look determines the decision cost; later looks serve validation. A deterministic support bound can produce a genuine zero-path decision. Table 3 reports which monetary tolerances are certified and the sampling cost of reaching a decision.

Table 3: Certification of monetary shortfall tolerances

Policy	Shortfall	0.10 (%)	0.25 (%)	0.50 (%)	Mean paths	Charged CPU (ms)
$\theta = 100$	0.353489	0.00	0.00	100.00	5,600	1.230
$\theta = 110$	0.000139	100.00	100.00	100.00	0	0.796
$\theta = 120$	0.161905	0.00	100.00	100.00	596	0.917
$\theta = 130$	0.456311	0.00	0.00	82.03	32,480	2.529
Always wait	1.156991	0.00	0.00	0.00	32,768	2.734

Certification frequencies for the terminal-call assessor, across 256 independent repetitions. The simultaneous family contains five policies, three methods and five fixed looks, hence 75 allocated bounds; one bound serves all three tolerances. Paths and charged CPU at tolerance 0.50 include unresolved assessments. Costs add the assessment prefix to one frozen independent setup measurement, which is not rerun in each repetition. Shortfalls are display midpoints of certified intervals. Simulation and confidence-limit arithmetic remain floating-point calculations.

Both zero and stock assessments remain unresolved in every repetition at every tolerance. Their expected dual slacks are approximately 1.63327 and 1.55275, respectively. The fitted stock coefficient is 0.217694. Its cheaper control reduces slack slightly, but the remaining slack exceeds every tolerance. Increasing the evaluation sample cannot remove that obstacle. Terminal-call centring changes the assessment of the same policies: all acceptable combinations are certified in every repetition except $\theta = 130$, $\eta = 0.50$, which succeeds in 210 of 256 repetitions. Its true shortfall is 0.456311, leaving relatively little tolerance for sampling uncertainty.

No repetition falsely certifies an unacceptable policy anywhere in the family. The 95 per cent Wilson interval for that family-event frequency is $[0, 1.48]$ per cent. This counts false acceptance at the selected tolerances; it does not measure coverage of every gap expectation. Gain evaluation, tail formulas and confidence-limit arithmetic remain floating-point calculations, distinct from the verified population reference.

The separately executable root-free bound calculation takes 0.762 CPU milliseconds for the entire threshold family. Direct verified quadrature takes 10.837 CPU milliseconds for all five policies. These single-pass observations exclude process start-up, imports and serialisation. Each standalone simulation decision is charged its observed assessment prefix plus one retained setup measurement; finite-threshold terminal-call assessments conservatively charge the complete analytical family calculation. This is phase-cost accounting, rather than independently repeated set-up timings, and supplies no universal processor ranking. In this tractable model, verified quadrature resolves the complete decision family directly. Simulation has to earn its relevance in the multidimensional setting, where its alternatives and numerical accuracy must also be compared.

5 Multidimensional policy performance and numerical references

The ten-case matrix uses 50,000 training paths and 200,000 independent evaluation paths per case, with 20,000-path chunks. I retain the random-stream layout across cases, so their estimates are not independent replications of a general claim. Policy value, expiry-only value and their matched difference are all reported in present-value units. The polynomial policy uses 6, 10

and 21 basis functions in two, three and five dimensions, respectively. Table 4 reports the policy rewards and their matched gains over waiting. Figure 3 shows how those gains vary with dimension and initial moneyness within the benchmark matrix.

Table 4: Independent max-call policy evaluation

Assets	Spot	Policy value	SE	Expiry gain	Gain SE
1	100	7.91930	0.02640	1.90917	0.02762
2	90	8.04852	0.02760	1.36918	0.02435
2	100	13.83711	0.03433	2.61931	0.03397
2	110	21.27218	0.03986	4.32145	0.04382
3	90	11.26762	0.03195	1.70464	0.02746
3	100	18.62946	0.03938	2.92325	0.03656
3	110	27.47272	0.04544	4.31120	0.04514
5	90	16.58460	0.03717	1.99711	0.03068
5	100	26.07060	0.04444	3.03087	0.03861
5	110	36.66863	0.05039	4.01621	0.04536

Present-value units. Each fixed degree-two policy uses 50,000 training paths and 200,000 independent evaluation paths. Expiry gain is a matched-path comparison with always waiting. Cases share a seed layout.

Every matched-gain interval over expiry-only exercise is positive. In the five-asset at-the-money case, the fitted policy value is 26.070601, its standard error is 0.044436 and the matched gain is 3.030866 with standard error 0.038613. The gain is economically interpretable within the model: it is the expected reward added by the fitted early-exercise decisions over waiting, rather than a percentage reduction in an auxiliary gap. The published five-asset at-the-money interval is $[26.109, 26.292]$; the present policy interval $[25.983507, 26.157694]$ overlaps part of it. That comparison supplies context, rather than an estimate of optimality or a certified shortfall, because the published algorithm, basis and simulation budget differ.

The expiry-only calculation supplies an independent implementation check. For independent assets, the maximum's CDF is the product of its marginal CDFs, so its European value reduces to a one-dimensional survival integral. Terminal Monte Carlo means differ from those quadrature values by at most 0.742 estimated standard errors across the matrix. This check addresses simulation and discounting, rather than optimal stopping.

The two-asset diagnostic plots discounted reward minus fitted continuation at $t = 1$. Figure 4 uses the full retained grid to locate the sign changes of the fitted exercise margin. Colour encodes its time-zero monetary magnitude, and the black curve interpolates the zero contour. Positive margins select exercise; ties continue. The dependence on the competing asset is visible at a fixed maximum. This is a diagnostic of the fitted rule and its polynomial extrapolation.

For the one-asset ten-right case, a Crank–Nicolson PDE with Rannacher smoothing and exercise projections at the retained dates gives 7.96377209 at 3,200 space intervals and 3,600 time steps on $[0, 400]$. The last joint-refinement change is 5.85×10^{-5}; doubling the finest time count changes the value by 7.32×10^{-7}. Larger domains are compared at unchanged spatial spacing. The policy estimate 7.919297 has standard error 0.026405. The difference from the PDE point reference is 0.044475, so ordinary reward sampling alone is insufficient to resolve the policy shortfall precisely. Grid changes are numerical diagnostics, not certified truncation errors. Figure 5 places reference refinement beside the policy's sampling interval, separating numerical resolution from uncertainty in the estimated policy reward.

6 Verified conditional integration and gain refinement

For each $j = 1, \ldots, J$ and a frozen fitted function v_j, set $c_j = \mathbb{E}[v_j(S_j) \mid \mathcal{F}_{j-1}, \mathcal{D}]$. The exact-centred increment $v_j(S_j) - c_j$ defines a martingale irrespective of approximation quality, provided it is integrable. I fit value functions on a separate simulation stream using a constant, the mean

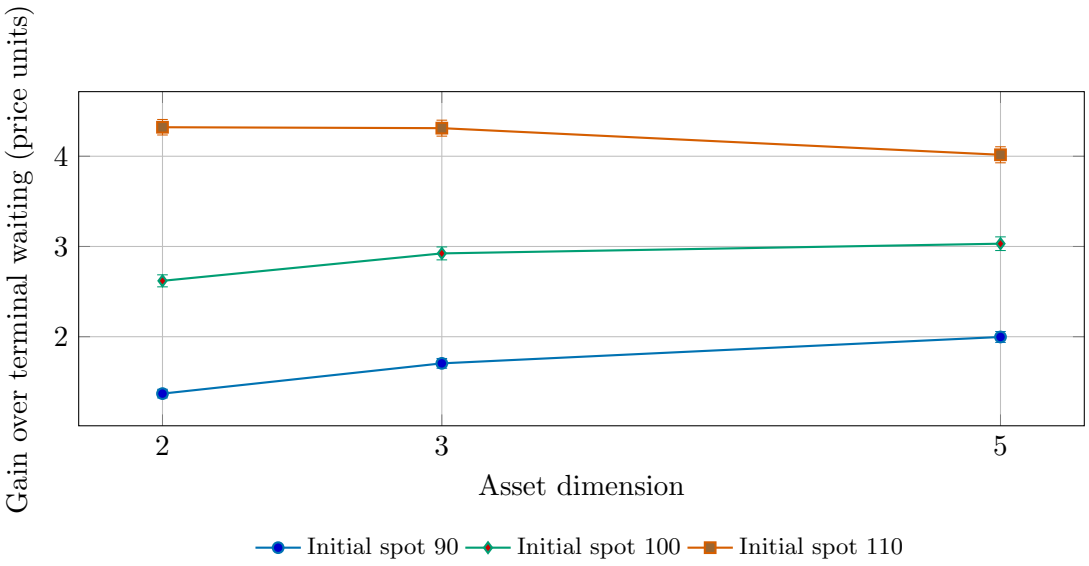


Figure 3: Matched exercise-policy gain for the public max-call process. Each degree-2 full-state polynomial policy uses 50,000 training paths and 200,000 independent evaluation paths. Bars are approximate pointwise 95% Student intervals conditional on the fitted policy. Cases share a seed layout; the intervals make no simultaneous or independent-case claim.

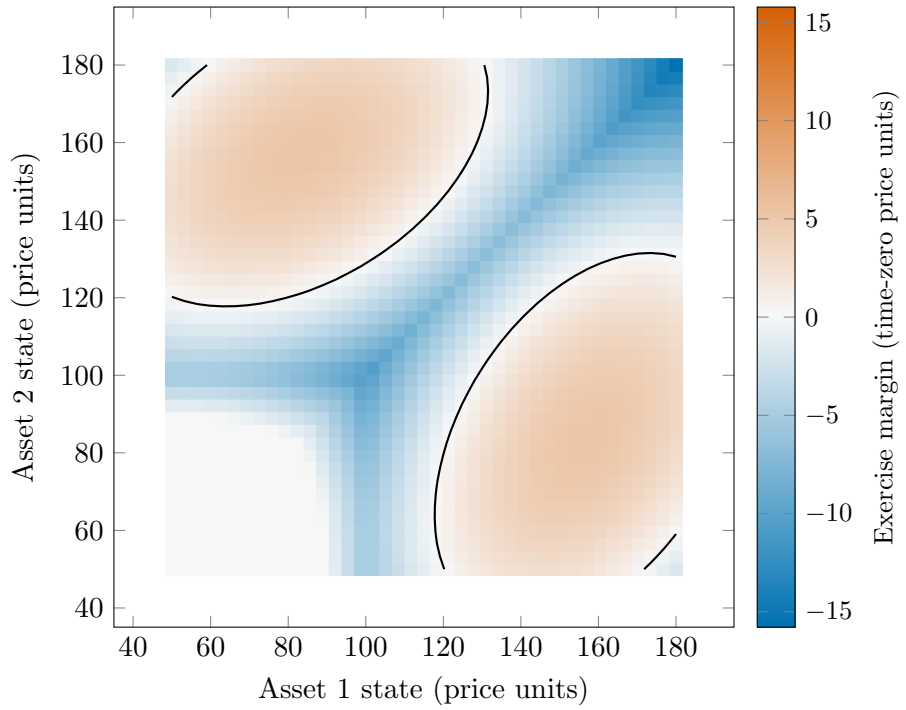


Figure 4: Fitted two-asset max-call policy margin at t=1 years, in time-zero price units. Positive margins favour exercise; ties continue. The full saved 41 by 41 state grid and its zero contour are shown. This is a fitted-policy diagnostic, not an optimal exercise boundary; polynomial extrapolation limits apply.

stock, four maximum hinges and four averaged individual hinges at strikes 80, 100, 120 and 140. These symmetric features deliberately restrict the value approximation; the exercise policy retains the sorted full state. The terminal hinge is payoff-scaled using the retained binary discount coefficient; the hybrid benchmark below instead encloses the exact exponential model

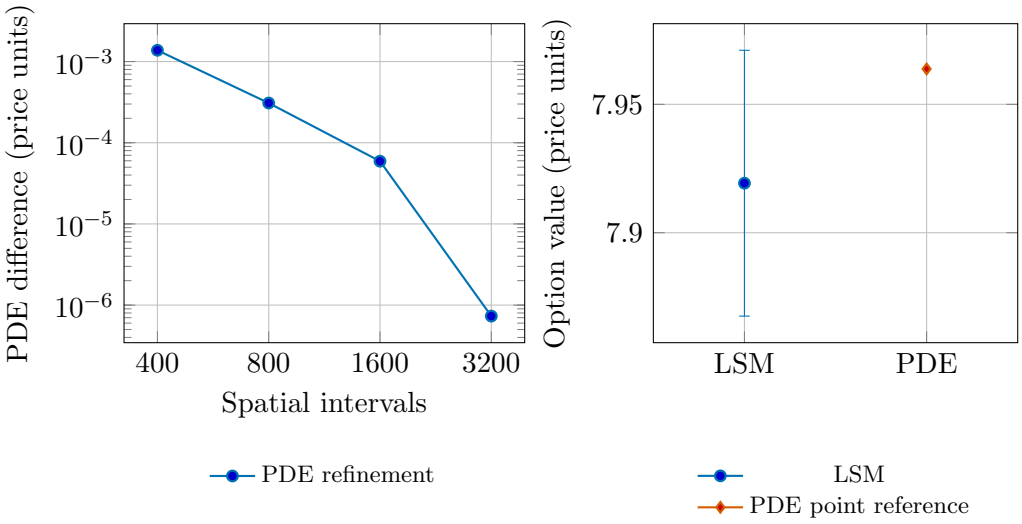


Figure 5: One-asset public-model reference. Left: joint spatial/time refinement compared with the finer-time PDE point reference. Right: the fitted policy value and its approximate pointwise 95% sampling interval beside that PDE value. These numerical convergence checks are not certified price or loss enclosures.

discount. Least-squares value fits use the SVD minimum-norm solution, retaining the effective rank and condition diagnostics. An incomplete empirical rank restricts the fitted approximation; it does not invalidate exact centring of the frozen function.

For a maximum hinge $\psi_k(x) = (\max_i x_i - k)^+$ under independent homogeneous GBM with $\sigma > 0$, $k > 0$ and $\Delta t > 0$,

$$\mathbb{E}[\psi_k(S_j) \mid S_{j-1} = x] = \int_k^\infty \left\{ 1 - \prod_{i=1}^{d} \Phi\left(\frac{\log(z/x_i) - (r - q - \sigma^2/2)\Delta t}{\sigma\sqrt{\Delta t}} \right) \right\} dz. \tag{12}$$

This established survival-integral representation uses all current coordinates in the conditional calculation. Independence is essential to the product formula and reduces each maximum-hinge expectation to a one-dimensional integral. The computational question concerns the number and required accuracy of these conditional queries; it is not an irreducible five-dimensional integration problem. I evaluate the finite integral and its analytical lognormal tail with outward arithmetic, following Johansson (2017) and Johansson (2018). Appendix B derives the tail allowance. Increasing precision and the cutoff controls different sources of error; the actual enclosure width determines when refinement ends.

6.1 Signed error propagation

Let $\widehat{c}_j$ be a predictable numerical centre and set $A_j = \sum_{r=1}^{j} \{v_r(S_r) - \widehat{c}_r\}$. The true martingale satisfies $A_j = M_j + B_j$, where B_j accumulates the centring errors $\mu_r = c_r - \widehat{c}_r$. For the selected date τ, let $[L_j, R_j]$ enclose $B_j - B_\tau$ using only the errors between dates j and τ, reversing signs for earlier candidate dates. With $z_j = H_j - A_j - H_\tau + A_\tau$,

$$\max_j (z_j + L_j) \le G_\pi(M) \le \max_j (z_j + R_j). \tag{13}$$

The shared increments cancel, so the bound pays only for the span that can change a candidate relative to the policy's choice. This is the structural reason to direct refinement towards those spans. Appendix B.2 gives the complete definitions and derivation.

The frozen hinge functions and finite-horizon GBM have the required finite moments. Predictable centres inside uniformly finite-width expectation brackets are integrable, so the true martingale retains the optional-sampling interpretation.[3]

6.2 A refinement experiment

I first obtain coarse expectation brackets and record their centres. I then compare uniformly tighter conditional brackets with refinement of increments that influence the current gain interval. Adaptive refinement retains its initial coarse centres; cold uniform evaluation selects centres from its fine brackets. Both enclose the same exact-centred gain with the policy and fitted value functions fixed. Candidates already dominated by the current lower maximum need no refinement. A remaining candidate depends only on increments between its date and τ; these intervals determine which conditional queries can narrow its bound. Post-processing may inspect the complete path because it encloses the same fixed functional rather than changing the exercise rule.

The comparison charges the initial brackets, unsuccessful accuracy attempts, tail calculations, arithmetic and refined queries. A numerical-width target concerns the realised gain, not the sampling error of its mean. If the target is unresolved, the wider interval is retained or the whole calculation fails explicitly; difficult paths are never silently removed. Table 5 compares conditional-query work across the three required widths; its final row reverses the strategy order to check whether the observed saving depends on execution order.

Table 5: Conditional-integration cost at a required maximum gain width

ε	Uniform calls	Refined calls	Uniform CPU	Refined CPU	Refined width
0.1	15490	9104	15.796	7.963	0.0997
0.01	16336	9506	17.040	8.885	0.00729
0.001	16645	9570	17.868	8.728	0.000404
0.01	16336	9506	16.516	8.626	0.00729

Both strategies meet the stated maximum path-gain width. Achieved widths can differ. Uniform costs exclude unnecessary coarse initialisation; adaptive costs include every coarse and refined query. Preparation and decision checks are reported separately. Calls count finite maximum-hinge integrals; CPU is measured in seconds.

I use a fresh five-asset, at-the-money fit with 50,000 policy-training paths, 50,000 separate value-fitting paths and 256 independent evaluation paths. The root seed is 2026100408. The design retains targets $\varepsilon \in \{0.1, 0.01, 0.001\}$ before assessment. The adaptive procedure starts with conditional widths at most one; the uniform procedure starts directly at ε/J and is charged no unnecessary coarse initialisation. Both meet the prescribed maximum gain width on every path, although uniform refinement often attains substantially narrower widths.

At $\varepsilon = 0.01$, conditional-integration calls fall from 16,336 to 9,506. Certificate-phase CPU falls from 17.040 to 8.885 seconds. Reversing the strategy order produces the same certificates and counts, with 16.516 and 8.626 seconds respectively. These repeated timings check an ordering effect on this machine; they are not a confidence interval for transferable performance. Policy checking and preparation are charged separately. Preparation takes 6.348 CPU seconds, and the standalone preparation-plus-assessment budgets at this target are 23.540 and 15.370 seconds. Across all four target passes, charging preparation once gives 74.160 seconds for uniform refinement and 41.126 seconds for adaptive refinement. [4]

At $\varepsilon = 0.001$, adaptive refinement gives the mean gap enclosure $[0.3525784484, 0.3525932414]$ to the displayed precision. Its midpoint sample mean has an ordinary estimated standard error of 0.058912, compared with numerical half-width 7.40×10^{-6}. The arithmetic enclosure is

[3] Optional sampling gives a zero stopped expectation before conditioning on a particular stopping event. It does not imply $\mathbb{E}[M_\tau \mid \tau = j] = 0$.

[4] These computational timings exclude environment start-up and JSON input/output.

therefore much narrower than the sampling uncertainty. Additional conditional precision alone will have little effect on the statistical assessment at this path count. The interval encloses the sample-average functional, rather than the population mean or the policy shortfall.

The certificate covers real conditional expectations and real gain algebra at supplied exact binary states, dates and coefficients. Fitted-function evaluation uses rational arithmetic, discount factors use Arb and exported endpoints retain rational numerator/denominator pairs. A separate check verifies the observed stopping decisions against the frozen adapted polynomial rule. This does not certify that a floating-point simulator samples the ideal GBM law or that a statistical interval covers its expectation. Numerical and statistical guarantees therefore remain separate.

6.3 Repeated fits and exact-moment benchmarks

I repeat the comparison on eight independently fitted five-asset policies. The root seeds run from 2026100410 to 2026100417. Every fit has 50,000 policy-training paths, 50,000 separate value-fitting paths and 256 independent outer paths. I fix the target $\varepsilon = 0.01$ before training and alternate the order of the refinement strategies, polynomial degrees and verified benchmark arms. The independently fitted policy and value-function pair is the replication unit. Within-fit paired standard errors describe conditional evaluation noise; across-fit standard errors use the eight fitted-model means.

I compare the hinge martingale with complete degree-two and degree-three polynomials in the unsorted full state, using the same value-fitting stream and exercise policy. The established regression-later construction of Glasserman and Yu (2004) permits analytical conditional moments. Appendix C derives their conditional moments from the Gaussian transition. These analytical centres remove conditional quadrature from the polynomial benchmark. A second benchmark retains the same interior polynomials and uses the exact terminal payoff $v_J = H_J$, isolating the error from fitting a polynomial to the terminal kink.

For fixed interior increments and the unchanged policy, conditional Jensen makes the exact-terminal hybrid's expected gap weakly smaller. This ordering is in expectation. The derivation and its integrability conditions are in Appendix C.

For the same policy, subtracting equation (5) gives

$$\mathbb{E}[G_\pi(M^p) - G_\pi(M^h) \mid \mathcal{D}] = s_{M^p} - s_{M^h}. \tag{14}$$

The policy shortfall cancels. I use matched path differences to compare expected dual slack without requiring the unknown five-dimensional optimum. A smaller expected gap then means a tighter upper valuation of the unchanged policy. This identity supplies the financial interpretation of the comparison. Table 6 compares the resulting gap quality and complete computational cost across all eight fits, including the exact-moment alternatives.

Adaptive refinement uses a median 59.1 per cent of the uniform integration calls, with fit ratios between 58.0 and 60.2 per cent. Its certificate-phase CPU ratio has median 51.9 per cent and range 50.2 to 53.1 per cent. Charging preparation and complete assessment raises the median ratio to 53.8 per cent, with range 52.1 to 54.8 per cent. Both strategies attain the prescribed target on all 2,048 retained paths, and every paired path enclosure overlaps. Uniform maximum widths range from 2.22×10^{-4} to 1.21×10^{-3}; adaptive maxima range from 0.00595 to 0.00981. The saving concerns meeting a common required tolerance, with visibly different achieved precision.

The adaptive mean gap across fits is 0.4055, with an across-fit standard error of 0.0248. The strongest polynomial comparator in this experiment is the degree-three hybrid: its mean is 1.2213, and its paired excess over the adaptive hinge is 0.8158 with across-fit standard error 0.0438. All eight paired fit means are positive, ranging from 0.5999 to 1.0126. The arithmetic half-width of their mean difference is 8.62×10^{-5}, much smaller than the sampling diagnostic.

Table 6: Repeated-fit verified dual assessment

Assessment	Mean gap	Fit SE	Δ (fit SE)	CPU (s)
Hinge, uniform	0.4055	0.0248	–	17.074
Hinge, adaptive	0.4055	0.0248	–	9.196
Degree two, polynomial	2.3749	0.0773	1.9694 (0.0870)	2.062
Degree two, hybrid	1.7049	0.0561	1.2994 (0.0617)	2.530
Degree three, polynomial	1.7780	0.0541	1.3725 (0.0684)	4.798
Degree three, hybrid	1.2213	0.0349	0.8158 (0.0438)	4.916

Eight independent fits of the five-asset benchmark use 50,000 policy-training paths, 50,000 value-fitting paths and 256 independent outer paths each. Every assessment attains maximum path-gap width 0.01. Fit SE uses eight fitting-unit means. The paired difference is comparator minus adaptive hinge for the same frozen policy and measures relative expected dual slack, not policy improvement. CPU is the median complete standalone budget including preparation and policy validation; hybrid setup uses measured component costs. Arithmetic sample-mean enclosures and within-fit diagnostics are retained separately.

These observations support a difference in upper-bound quality; they do not identify either method's absolute slack or the optimal policy value.

The degree-three hybrid costs a median 4.916 CPU seconds, compared with 9.196 seconds for adaptive hinge assessment and 17.074 seconds for uniform hinge assessment. Thus adaptive refinement reduces the cost of the tighter hinge assessment, while the moment benchmark remains cheaper. The complete cost includes common policy and outer-path preparation, each method's own value-fitting work, verified assessment and policy checks. Hinge regression is excluded from polynomial budgets. Hybrid setup is reconstructed from measured interior-regression components rather than independently rerun. Each verified arm owns its moment-factor cache and is charged for its construction. Process start-up and file serialisation are excluded. Figure 6 presents this quality and cost trade-off.

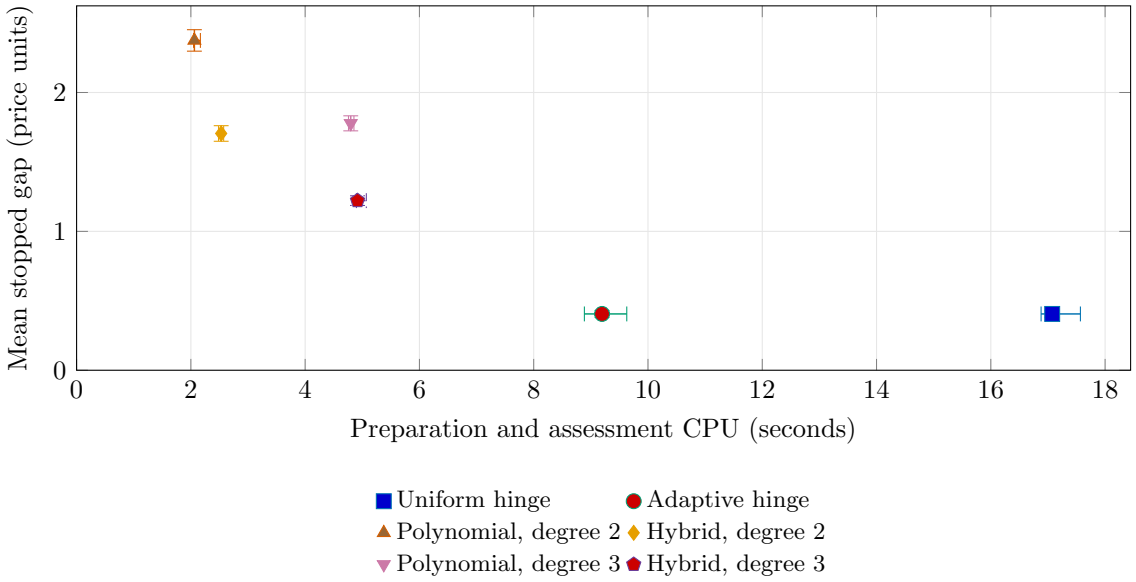


Figure 6: Cost and quality of verified dual assessments across eight independent fits. Points show the median preparation and assessment budget and mean stopped gap. Horizontal bars span observed minimum and maximum CPU; vertical bars give one across-fit standard error of the gap mean. These ranges and standard errors are not confidence intervals. Budgets include policy validation. Process start-up and result serialisation are excluded. Every method attains maximum path-gap width 0.01.

6.4 Independent larger-sample evaluation

I next assess the eight retained fits on 1,024 fresh paths each, using a new simulation stream independent of training, value fitting and the earlier 256-path evaluation. I preserve every policy coefficient, hinge function and degree-three polynomial coefficient. The hybrid was selected as the strongest polynomial comparator in the earlier experiment; the follow-up tests that fixed choice on new observations. All three assessors share the new paths within each fit. I fix their execution order before sampling, using all six permutations followed by two rotations. Each assessor occupies each position two or three times, and uniform evaluation precedes adaptive evaluation in four fits and follows it in four.

I condition on the retained fits and earlier comparator selection, denoted by $\mathcal{D}_0$. The conditional standard error combines the within-fit variances of the independent fresh-sample paired means. Across-fit dispersion also includes differences between population gaps and cannot isolate that evaluation uncertainty. Appendix C.2 gives the calculation. The frozen functions have finite second moments. Table 7 reports the sampling diagnostics alongside the separate arithmetic mean enclosures and computational costs.

Table 7: Fresh-sample assessment of retained five-asset fits

Assessment	Mean gap	Conditional SE	Fresh CPU (s)	Standalone CPU (s)
Hinge, uniform	0.3719	0.0105	73.336	73.872
Hinge, adaptive	0.3719	0.0105	38.512	39.023
Degree three, hybrid	1.2552	0.0222	13.237	15.034

Each of eight previously trained fits is held fixed and assessed on 1,024 fresh outer paths. The degree-three hybrid was selected from the earlier comparison; its new evaluation stream is independent of that sample. Across-fit dispersion is retained separately as a descriptive measure, not a confidence interval. Conditional SE is the square root of the sum of squared within-fit standard errors, divided by eight. The paired hybrid-minus-adaptive mean is 0.8833, with conditional SE 0.0233 and descriptive across-fit SE 0.0184. Both CPU columns report medians over eight fits. Fresh CPU includes new outer preparation and the complete assessment with policy validation. Standalone CPU reconstructs a budget by adding retained fitting and setup components to that fresh work; the original outer preparation is replaced and fitting is not rerun. All arms attain maximum path-gap width 0.01. Paired differences identify relative dual slack for the same frozen policy.

On the new sample, the adaptive hinge mean is 0.3719 and the hybrid mean is 1.2552. Their paired difference is 0.8833, with conditional standard error 0.0233; its numerical half-width is 7.98×10^{-5}. The descriptive across-fit standard error is 0.0184. Equation (14) interprets the paired difference as relative expected dual slack for the same policies. The hinge assessment requires a median 39.023 standalone CPU seconds, compared with 15.034 for the hybrid. Its tighter mean gap therefore carries an explicit computational premium. The exercise decisions remain unchanged.

Both hinge strategies meet the required width on all 8,192 new paths, and their paired path enclosures intersect throughout. Against the original ε/J uniform rule, adaptive refinement uses a median 59.2 per cent of integration calls and 52.8 per cent of the reconstructed standalone CPU budget. Fresh CPU includes new outer simulation, frozen-policy evaluation and complete assessment. The standalone budget adds the appropriate retained fitting measurements, replacing the old outer preparation and excluding all earlier assessments. No fitting is rerun. The comparison extends the computational evidence to a larger independent sample. Section 6.5 then tests whether the saving persists against stronger common-tolerance controls on those same paths.

6.5 Common-tolerance controls

The cold uniform rule above requests conditional widths of ε/J and often achieves much narrower gain intervals than required. I therefore compare two controls that stop using the actual gain

width. The first starts at a common conditional tolerance $\theta_0 = \varepsilon$; the second starts at $\theta_0 = 1$, matching adaptive initialisation. Each fixes its own initial predictable centres. If the gain width remains unresolved, I quarter one common tolerance and query every conditional bracket whose retained width exceeds it. I intersect the new brackets with their predecessors, complete the whole round and then test the gain. Already compliant brackets are retained. Neither control ranks dates or uses candidate dominance to choose its queries.

For $J \geq 1$, let $w(I)$ denote the width of an interval and let I_{H_j} enclose the discounted reward at date j. If every conditional bracket has width at most θ, define

$$\rho_\tau = \max_{j \neq \tau} \{w(I_{H_j}) + w(I_{H_\tau})\}, \qquad L_\tau = \max(\tau, J - \tau). \tag{15}$$

The shared-increment gain enclosure I_G satisfies

$$w(I_G) \leq \rho_\tau + L_\tau \theta. \tag{16}$$

Cancelling shared increments leaves at most L_τ expectation brackets and the reward widths. Appendix B.3 gives the full argument. If $\rho_\tau < \varepsilon$, a common tolerance $\theta \leq (\varepsilon - \rho_\tau)/L_\tau$ suffices, but each control stops using the actual gain interval rather than that sufficient condition. I retain the sixteen-round limit and every unresolved target or conditional-oracle failure. The absolute residual enclosure remains a separate cross-check.

I reuse the eight retained 1,024-path inputs and rerun adaptive assessment alongside the two controls. This is a matched numerical comparison, with no new fitting or evaluation sample. The newly incurred assessment budget charges each complete assessment. A reconstructed standalone budget adds the retained input preparation charge once. Complete timings include each implementation's result construction; the common controls retain fuller query traces. Integral and integrand-evaluation counts measure numerical work separately from these implementation costs.

Table 8 tests whether common tightening explains the saving against the original conservative rule. The coarse-start control is the cheaper of the two common-tolerance implementations. Relative to that control, adaptive refinement uses a median 78.8 per cent of conditional-integral calls and 75.7 per cent of reconstructed standalone CPU. The paired CPU ratios range from 66.9 to 78.3 per cent, below one in every fit. These are medians of within-fit ratios, rather than ratios of the table's column medians. All three methods attain the required width on every path; their largest actual width is 0.0099548. Each new interval intersects the other two and the retained conservative-uniform interval, and adaptive numerical records reproduce the earlier assessment exactly.

Table 8: Uniform accuracy budgets and gain-directed refinement

Assessment	Integral calls	Maximum width	New CPU (s)	Standalone CPU (s)
Cold common budget	495,363	0.0099548	65.920	66.436
Coarse-centre common budget	392,680	0.0099548	47.811	48.353
Gain-directed refinement	308,308	0.0099548	36.502	37.019

All three arms assess the same 1,024 retained paths per fit across eight fixed fits and attain required maximum path-gap width 0.01. Integral calls total all eight fits; maximum width is the largest actual width across 8,192 paths. CPU columns give medians over eight fits. New CPU includes the complete assessment, policy validation and result construction. Standalone CPU adds retained preparation once, without rerunning simulation or fitting. Common controls retain fuller query traces; measured CPU compares verified implementations and does not isolate allocation-rule runtime.

The saving remains after the controls stop using actual gain width, but is smaller than the earlier conservative-baseline comparison. The result supports selective refinement for these fixed functions and inputs. It compares complete verified implementations, including their different trace construction. Measured costs cover successful assessments from two sessions;

three continued fits ran in separate processes. The ratios compare methods within each fit. The comparison supplies no additional evaluation sample or coverage experiment.

I also test allocation for the direct European-value martingale on the first 32 retained paths of each fit. With $P_j = \mathbb{E}[H_J \mid \mathcal{F}_j, \mathcal{D}]$ and $F_j = H_j - P_j$, its restricted gain is

$$G_\pi^{\mathrm{E}} = \max_{1 \le j \le J} (H_j - P_j) - (H_\tau - P_\tau). \tag{17}$$

The terminal level F_J is zero, and the candidate at τ is zero after subtracting F_τ. Both rules share structural screening, safe dominance pruning, initial price-width target one, gain-width target 0.01 and the same query-record fields. Common refinement uses 1,934 primitive integrals and adaptive refinement 1,924; every gain interval meets the target and intersects a separately recomputed bracket of width at most 2.74×10^{-9} from the same pricing routine. The median paired assessment CPU ratio is 98.5 per cent, ranging from 96.0 to 101.0 per cent. Each fitted-unit assessment takes approximately 0.25 to 0.28 seconds. Shared initial queries account for 90.8 per cent of the common rule's integrals, and 186 of 256 paths need no further refinement. This comparison provides little evidence of a useful allocation advantage for the European representation. I retain common European refinement as a simpler comparator in the monetary decision experiment below; I do not extrapolate the hinge result to every assessment martingale.

6.6 Removing a dominated initial right

A monetary assessment should not pay for a martingale increment that cannot affect its retained candidates. In the five-asset study, $S_0^i = K = 100$, so $H_0 = 0$, whereas every later reward is non-negative. The strict exercise rule never stops at zero. The following calculation identifies the restricted evaluator that I use for a prospective tail bound.

I set $N_0 = N_1 = 0$ and use the exact-centred increments from date two onwards. The restricted gain is

$$N_j = \sum_{r=2}^{j} \{v_r(S_r) - c_r\} \quad (j \ge 2), \qquad G_\pi^{\mathrm{r}} = \max_{1 \le j \le J} \{H_j - H_\tau - (N_j - N_\tau)\}. \tag{18}$$

Proposition 1 in Appendix E proves that the optimal stopping value is unchanged and that

$$0 \le G_\pi^{\mathrm{r}} \le G_\pi \quad \text{almost surely}, \qquad \mathbb{E}[G_\pi^{\mathrm{r}} \mid \mathcal{D}] \ge \ell_\pi. \tag{19}$$

The proof derives admissibility and the zero stopped expectation under the pricing law conditional on a fixed fit.

I use this application of the existing dual bound because every retained candidate-policy difference omits the first fitted increment. The calculation changes the evaluator while preserving the policy and the optimal stopping value. Integrability is required under the fixed-fit pricing law; no integrability of random regression coefficients over the training procedure is asserted.

All preceding comparisons retain their original full calendar. In a separate exact reconstruction from the coarse-common traces, the upper endpoint of candidate zero lies strictly below the lower maximum over retained candidates on all 8,192 paths. The restricted and full signed intervals are identical on every path, so the observed mean reduction is exactly zero. This reconstruction requires neither new paths nor conditional integrations. Deleting a candidate need not reduce an enclosure's width, although every reconstructed interval here still meets 0.01. The first value-function increment disappears algebraically; the remaining extrapolation tail must still be bounded before the observed mean can support a finite-sample monetary decision.

6.7 Monetary assessment of frozen multivariate policies

A small numerical enclosure does not by itself decide whether a policy is acceptable. I therefore freeze two upper assessors before the production evaluation of each of the eight five-asset

coefficient sets: the zero martingale and the direct European-value martingale. Their gains are $G_\pi^0 = \max_{1\le j\le J} H_j - H_\tau$ and $G_\pi^{\mathrm{E}} = \max_{1\le j\le J} F_j - F_\tau$, respectively. Both dominate the policy shortfall in conditional expectation. They assess the same exercise decisions, but their dual slack and computational costs can differ substantially.

I fix the tolerances at 0.5 and 1 option-price unit, or 0.5 and 1 per cent of strike. Each fit receives 16,384 costed paths, with looks at 4,096 and 16,384. A provenance-based eligibility amendment leaves fit 1 with 4,080 and 16,368 observations for inference; the other fits retain both complete prefixes. Appendix D records the deviation and its full cost treatment. Caps, tolerances, probability allocation, coefficients and the strict exercise convention remain unchanged. The assessed implementation is an adapted Borel own-state rule; Appendix D.2 specifies its numerical decisions and failure handling.

The clipping caps are 150 for G_π^0 and 50 for G_π^{E}. Appendix F derives their evaluator-specific tail allowances and the variance enclosure. With n eligible observations, clipped endpoint upper mean $\bar{u}$, midpoint standard deviation s_m and radii r_i, I form

$$U_{a,n} = \bar{u} + \left(s_m + \sqrt{\frac{\sum_i r_i^2}{n-1}} \right) \sqrt{\frac{2\log(2/\delta)}{n}} + \frac{7B_a \log(2/\delta)}{3(n-1)} + T_{B_a}^a. \tag{20}$$

I allocate $\delta = 1/640$ to each of the $8 \times 2 \times 2 = 32$ upper events. The union bound supplies simultaneous conditional coverage of at least 95 per cent when the observations are iid under the stated pricing law and their numerical intervals contain the corresponding ideal gains. The five-asset implementation here encloses supplied binary-state calculations; it does not verify that observation premise for unrounded continuous paths. Matching assessors and nesting the looks do not require another independence assumption. The two tolerances interrogate the same upper events and consume no additional probability allocation. The first upper endpoint no greater than a tolerance determines its required prefix; a nondecision retains the full maximum budget. A nondecision is not evidence that the policy's actual shortfall exceeds the tolerance.

Required-prefix CPU charges combine measured path preparation, each assessor's own construction and evaluation, numerical and evidence work, shared verification, and retained policy training once. Value fitting and earlier assessments are excluded. Since preparation completes before assessment, these are reconstructed standalone prefix charges. The after-run eligibility correction is charged separately and conservatively added in full to each corrected required-prefix comparison. Actual newly executed workflow and independent verification costs remain separate. Outward arithmetic certifies quantities at supplied binary states; it does not enclose unrounded ideal states or certify the floating-point Gaussian simulator as an exact draw from the continuous model.

Table 9 reports the computed limits for each policy. No upper endpoint is at most one unit at the first look. At the final look, the European limits range from 0.9108 to 0.9486, all below one unit. The zero-martingale limits range from 12.1971 to 12.5154, above both tolerances. Neither assessor produces a limit at most half a unit. The thresholds are applied per policy, rather than to an average across fits. A simultaneous policy guarantee additionally requires the ideal-sampling and observation-enclosure premises stated above.

The original reconstructed European charges range from 187.79 to 244.94 CPU seconds per policy, compared with 15.67 to 17.93 for the zero assessor. The full after-run correction costs 434.84 CPU seconds. Adding it conservatively to each standalone decision gives European charges of 622.63 to 679.78 seconds. I report the original and correction charges separately because the correction reconstructs an already completed experiment. The complete newly executed preparation, assessment and correction workflows consume 2,186.47 CPU seconds across the family. Independent reconstruction checks have separate verification costs.

The assessment representation changes whether a computed limit lies below the tolerance. The European martingale is more expensive, but all eight of its final limits are below one unit;

the zero construction reaches neither tolerance within the declared budget. Under the ideal-sampling and observation-enclosure premises, this comparison supports a conditional monetary conclusion. Its common refinement rule suffices for the numerical calculation, and the earlier European allocation comparison provides little reason to use selective refinement here. The larger saving for fitted hinge functions remains specific to that representation; it has not been shown to lower the complete cost of a monetary assessment.

Table 9: Monetary assessment of the frozen five-asset policies

Fit	$\overline{G}^{\mathrm{E}}$	U^{E}_{4k}	U^{E}_{16k}	U^{0}_{16k}	EU cost (s)	Zero cost (s)
1	0.7466	1.0839	0.9252	12.4484	192.62	16.43
2	0.7350	1.1016	0.9134	12.4595	198.21	15.72
3	0.7469	1.0826	0.9264	12.3222	187.79	15.71
4	0.7498	1.1174	0.9280	12.5154	191.02	16.25
5	0.7686	1.1783	0.9486	12.3413	211.97	15.70
6	0.7324	1.1299	0.9108	12.1971	194.50	16.00
7	0.7455	1.0749	0.9244	12.4475	244.94	15.67
8	0.7370	1.0900	0.9158	12.2178	221.15	17.93

The first two upper-limit columns use the costed looks of 4,096 and 16,384 paths. Fit 1 excludes the sixteen exposed rows from inference, giving eligible counts 4,080 and 16,368. Other fits retain all rows. Mean gains are interval midpoints. CPU columns retain reconstructed original standalone charges, including all generated and assessed paths and policy training once. The complete after-run correction adds 434.84 CPU seconds to every reported decision charge; independent verification costs are separate. Acceptance uses exact upper endpoints rather than their rounded table displays. Limits share a simultaneous 95 per cent conditional guarantee under the stated ideal sampling and observation-enclosure premises. Both assessors evaluate unchanged policies; unresolved outcomes do not establish excessive actual shortfall.

7 Interpretation and remaining research tests

The monetary decision diagnostic shows when an assessment construction prevents a useful conclusion. Dual slack larger than the financial tolerance cannot be removed by increasing the evaluation sample. A tractable deterministic reference should be used where it is available; changing the evaluator becomes worthwhile when it permits a decision that the cheaper construction cannot reach.

The coverage diagnostic changes the operational reading of a small reported gap. A tight evaluation martingale does not ensure a reliable upper confidence limit. Conversely, a loose martingale can appear safe because its slack covers an inferential error. The clipped finite-sample calculation supplies a useful response in a setting with a tractable analytical tail. Its conservatism is visible and measurable, rather than hidden in an asymptotic approximation.

The multidimensional experiments show a computational saving alongside a cost and bound-quality trade-off. Gain-directed refinement meets the same realised gain-width requirement at a smaller reconstructed hinge-assessment budget across all eight fits, including the independent larger-sample evaluation. Stopping-aware common-tolerance controls reduce the apparent advantage, while cheaper exact-moment hybrids retain larger observed gaps. These are comparisons of the measured implementations and retained functions. In the separate monetary calculation, the European-value assessor produces final upper limits below one unit for every policy. The cheaper zero assessor reaches neither tolerance, and half-unit targets remain unresolved. Interpreting these limits as simultaneous continuous-model policy guarantees requires the stated sampling and ideal-observation enclosure premises. The result therefore identifies a useful evaluator trade-off under explicit assumptions; it does not establish that gain-directed refinement is the cheapest route to monetary acceptance. Broader models and accuracy-competitive alternatives remain tests of that further claim.

A Diagnostic reference calculations

A.1 Conditioning the zero-martingale gain

I calculate both expectations by one-dimensional lognormal quadrature. For the zero martingale, conditioning on $S_1 = x$ gives a call-spread identity. If $h(x) = e^{-rt_1}(x-K)^+$ and $\mathcal{C}(x,k)$ denotes the undiscounted conditional call expectation at T, then

$$\mathbb{E}[G_0 \mid S_1 = x] = h(x) - e^{-rT}\{\mathcal{C}(x,K) - \mathcal{C}(x, K + e^{rT}h(x))\}. \tag{A.1}$$

I use this identity to integrate the zero-martingale target without a noisy inner sample. The analytical call formula follows the lognormal pricing calculation of Black and Scholes (1973). Quadrature splits at the strike and exercise boundary. The retained quadrature error estimates are about 10^{-12}, with an analytical omitted-tail expectation bound below 3×10^{-30}. These are numerical reference values, not interval-certified quadrature.

A.2 Threshold monotonicity and verified reference

Write $D(s) = H_1(s) - C_1(s)$. Let Φ be the standard normal distribution function and $d_+(s) = \{\log(s/K) + (r - q + \sigma^2/2)(T - t_1)\}/\{\sigma\sqrt{T-t_1}\}$. For $s > K$ and $q > 0$,

$$D'(s) = e^{-rt_1}\{1 - e^{-q(T-t_1)}\Phi(d_+(s))\} > 0. \tag{A.2}$$

There is a unique exercise boundary $b > K$.

At the strike, $D(K) = -C_1(K) < 0$ in the stated strictly positive-volatility model. Above the strike, the derivative is at least $e^{-rt_1}(1 - e^{-q(T-t_1)}) > 0$. It follows that $D(s)$ tends to infinity and crosses zero exactly once. A threshold rule disagrees with the optimal rule only on the interval between θ and that crossing. Monotonicity therefore bounds its realised opportunity cost by $|D(\theta)|$ for finite $\theta \geq K$, without requiring the crossing to be calculated.

I separately enclose each population shortfall by verified integration of the signed opportunity cost between the threshold and the boundary. Strict interval signs bracket b within 5.56×10^{-15}. I integrate outside that uncertain strip and bound the strip contribution by

$$\frac{e^{-rt_1}(b_u - b_l)^2}{b_l \sigma \sqrt{t_1}\sqrt{2\pi}}. \tag{A.3}$$

The factor follows from $|D'| \leq e^{-rt_1}$ and the lognormal density bound on the strip. The always-wait integral additionally uses an analytical upper-tail allowance. Arb and Acb enclose arithmetic and integration error. Rational endpoints determine the tolerance decisions; the displayed decimals are summaries. This direct calculation resolves every policy/tolerance pair, including rejection of all three tolerances for always waiting.

B Conditional calculations and signed errors

B.1 Conditional maximum hinges

For the independent positive-volatility model in Section 6, the hinge is nonnegative and integrable. Tonelli's theorem gives its expectation as the integral of its survival probability. Conditional independence supplies the product of marginal lognormal distribution functions in (12). The missing integral beyond a cutoff is a maximum call with strike equal to that cutoff.

At cutoff $L > k$, the missing tail satisfies

$$0 \leq \mathbb{E}[(\max_i S_j^i - L)^+ \mid x] \leq \sum_i \mathbb{E}[(S_j^i - L)^+ \mid x]. \tag{B.1}$$

The individual tails have analytical lognormal expectations. I use verified Acb integration for the finite part and Arb arithmetic for the tails, following the established software and integration algorithms described by Johansson (2017) and Johansson (2018). Raising precision narrows arithmetic error; increasing L reduces the tail allowance. The actual enclosure width, rather than the requested integration tolerance, determines when refinement ends.

B.2 Signed centring errors

Let $\widehat{c}_j$ be a recorded numerical centre fixed from the preceding state. With $A_0 = M_0 = B_0 = 0$, define $A_j = \sum_{r=1}^{j}\{v_r(S_r) - \widehat{c}_r\}$ and $\mu_r = c_r - \widehat{c}_r$. Then $A = M + B$, where $B_j = \sum_{r=1}^{j}\mu_r$. Suppose $a_r \le \mu_r \le b_r$. For a realised policy date τ, define

$$L_j = \begin{cases} \sum_{r=\tau+1}^{j} a_r, & j > \tau, \\ -\sum_{r=j+1}^{\tau} b_r, & j < \tau, \\ 0, & j = \tau, \end{cases} \qquad R_j = \begin{cases} \sum_{r=\tau+1}^{j} b_r, & j > \tau, \\ -\sum_{r=j+1}^{\tau} a_r, & j < \tau, \\ 0, & j = \tau. \end{cases} \tag{B.2}$$

Writing $z_j = H_j - A_j - H_\tau + A_\tau$, I obtain

$$\max_j(z_j + L_j) \le G_\pi(M) \le \max_j(z_j + R_j). \tag{B.3}$$

I derive this formula by substituting the compensator into the stopped gain and cancelling increments shared by the candidate and selected dates. I use it because an all-absolute-errors allowance pays for increments that cancel. The inequality is an application of established duality and interval algebra, not a new general martingale theorem. The candidate $j = \tau$ is exactly zero.

Conditional on a finite frozen fit, the hinge functions have at most linear growth and finite-horizon GBM has finite moments. This supplies the required integrability. A numerical centre within a uniform finite-width bracket of c_j is also integrable. Optional sampling concerns the stopped expectation before conditioning on a particular stopping event; it does not imply $\mathbb{E}[M_\tau \mid \tau = j] = 0$.

The algebra in (B.3) follows from

$$H_j - M_j - H_\tau + M_\tau = z_j + B_j - B_\tau. \tag{B.4}$$

For $j > \tau$, summing $a_r \le \mu_r \le b_r$ over the forward span gives the displayed endpoints. For $j < \tau$, negating the reverse span exchanges its lower and upper endpoints. Taking the maximum preserves their order. At $j = \tau$, the shared expression is exactly zero.

B.3 A sufficient common-tolerance width

With the notation of Section 6.5, each nonselected candidate uses only the conditional brackets between its date and τ. All fitted values at supplied states are fixed within this conditional-error calculation; the reward enclosures contribute ρ_τ. Thus each candidate width is at most $\rho_\tau + |j - \tau|\theta$. The interval maximum has width no greater than the largest candidate width, which proves (16).

I obtain this sufficient bound by cancelling increments before $\min(j, \tau)$ in each stopped difference. The remaining span contains $|j - \tau|$ conditional brackets; the selected-date candidate is exactly zero. The width of the interval maximum is no greater than the largest candidate width. If $\rho_\tau < \varepsilon$, the condition $\theta \le (\varepsilon - \rho_\tau)/L_\tau$ is sufficient to meet the target. I use this calculation to explain how common tightening can resolve a gain certificate, rather than prescribing ε/J in advance. Each control stops using the actual interval, which may meet the target even when the sufficient condition fails. I allow at most sixteen global refinement rounds; unresolved targets and conditional-oracle failures are retained explicitly. The separate absolute residual enclosure is a cross-check and need not narrow when the initial centres remain fixed.

C Analytical comparators and conditional sampling precision

C.1 Polynomial centres and the exact-terminal hybrid

For a monomial with multi-index α, independence and the Gaussian transition give

$$\mathbb{E}\left[\prod_i (S_j^i/100)^{\alpha_i} \mid S_{j-1} = x\right] = \prod_i (x_i/100)^{\alpha_i} \exp\left\{\Delta t_j \sum_i \left[\alpha_i(r - q_i) + \tfrac{1}{2}\alpha_i(\alpha_i - 1)\sigma_i^2\right]\right\}. \tag{C.1}$$

I obtain this expression from the exponential moments of the independent normal increments. I use it to remove conditional quadrature from the polynomial benchmark. A second benchmark keeps its interior polynomials but sets $v_J = H_J$, using verified integration for the exact terminal max-call. This hybrid isolates the penalty incurred by fitting a polynomial to the terminal kink. Finite-degree polynomials have finite moments under finite-horizon GBM, so their exact-centred increments satisfy the same martingale and optional-sampling conditions as the hinge construction.

The exact-terminal hybrid is a stronger benchmark for fixed interior increments. Let $A = \max_{j<J}(H_j - M_j)$, $C = \mathbb{E}[H_J \mid \mathcal{F}_{J-1}, \mathcal{D}]$ and $Z_p = H_J - M_{J-1} - p(S_J) + \mathbb{E}[p(S_J) \mid \mathcal{F}_{J-1}, \mathcal{D}]$. Both terminal functions are integrable, and A is measurable at $J-1$. Conditional Jensen gives

$$\begin{aligned} \mathbb{E}[U_p \mid \mathcal{F}_{J-1}, \mathcal{D}] &= \mathbb{E}[\max(A, Z_p) \mid \mathcal{F}_{J-1}, \mathcal{D}] \\ &\geq \max(A, C - M_{J-1}) = U_{\text{hybrid}}. \end{aligned} \tag{C.2}$$

For the unchanged adapted policy, optional sampling then gives a weakly smaller expected gap. I use this standard argument to remove avoidable terminal regression error from the comparison; it implies no pathwise ordering and supplies no claim of a new theorem.

C.2 Conditional variance of the fresh-sample comparison

Let $\mathcal{D}_0$ contain the retained fits and the earlier comparator selection. For fit i, let $\widehat{\delta}_i$ be the mean of the matched hybrid-minus-hinge gaps, and let s_i^2 be the sample variance of those path differences. Under ideal independent sampling, the conditional variance is

$$\begin{aligned} \operatorname{Var}\left(\frac{1}{8}\sum_{i=1}^{8} \widehat{\delta}_i \,\middle|\, \mathcal{D}_0\right) &= \frac{1}{64}\sum_{i=1}^{8} \operatorname{Var}(\widehat{\delta}_i \mid \mathcal{D}_0), \\ \widehat{\mathrm{SE}}_{\text{conditional}} &= \frac{1}{8}\sqrt{\sum_{i=1}^{8} \frac{s_i^2}{1024}}. \end{aligned} \tag{C.3}$$

I obtain this expression by summing the conditional variances of the equally weighted independent estimates. I use it because dispersion between retained fit means also contains differences between their population gaps. That dispersion cannot isolate the uncertainty introduced by the fresh evaluation. The frozen functions have finite second moments under this finite-horizon GBM model. The midpoint standard errors are sampling diagnostics; the arithmetic mean enclosures remain separate.

D Protocol departures and implementation details

D.1 Exposed prefix and inference eligibility

Before registration, preparation fixtures had reused the first 16 rows of one policy's date streams. I discovered this overlap during execution and recorded an eligibility amendment before calculating corrected statistics. I preserve all rows and their costs, but exclude that prefix from

subsequent sampling inference. Its eligible counts are therefore 4,080 and 16,368; the remaining policies retain 4,096 and 16,384. The exclusion follows from stream provenance alone, with caps, tolerances and probability allocation unchanged.

The corrected statistics and costs in Table 9 retain this exclusion. The conditional argument in Appendix F concerns only the eligible suffix. The correction does not create a new sample or remove the computational expenditure of the excluded rows.

D.2 Implemented exercise rule

Separate per-date streams use keys $(4, j)$ under each retained fitting root, in batches of 256. The coefficients and strict exercise convention remain fixed. I evaluate each current-state polynomial in a fixed pathwise order, using products and `math.fsum`; ties and non-finite polynomial arithmetic continue, and surviving paths exercise at expiry. This removes any rounding dependence on the number of other surviving paths in a matrix product. For the ideal-path policy, each state is first rounded coordinatewise to the declared binary format; unrepresentable nonterminal decisions continue. This gives a total Borel own-state rule. It defines the assessed implementation rather than asserting exact real-polynomial signs or equivalence of every historical floating-point decision. Invalid simulated states fail the attempt instead of being replaced.

D.3 Historical market-data scope

I assess pricing and policy evaluation under stipulated processes. The historical data screen admitted no observations for an independent market comparison, so it supplies no test of model adequacy. Its records remain in the historical supplement.[5] The present research application must be supported by the numerical and decision experiments themselves.

E Dominated initial exercise right

Proposition 1 (A zero initial reward)**.** *Fix a finite fitted specification $\mathcal{D}$ and work under its conditional pricing law on the filtered space $(\Omega, \mathcal{F}, (\mathcal{F}_j)_{j=0}^{J}, \mathbb{Q})$, where $J \geq 1$ and the initial information contains the frozen fit. Suppose H_j is $\mathcal{F}_j$-measurable, non-negative and integrable at every date, with $H_0 = 0$ almost surely. Let $\mathcal{T}_{a:J}$ denote the set of all stopping indices taking values in $\{a, \ldots, J\}$, and let the assessed policy satisfy $\tau \in \mathcal{T}_{1:J}$. For integrable $\mathcal{F}_j$-measurable fitted values, set*

$$c_j = \mathbb{E}[v_j(S_j) \mid \mathcal{F}_{j-1}, \mathcal{D}], \qquad X_j = v_j(S_j) - c_j, \qquad M_j = \sum_{r=1}^{j} X_r. \tag{E.1}$$

For this M, let $G_\pi = \max_{0 \leq j \leq J}\{H_j - H_\tau - (M_j - M_\tau)\}$ be the full-calendar gain. With $M_0 = N_0 = N_1 = 0$, define

$$N_j = \sum_{r=2}^{j} X_r \quad (j \geq 2), \qquad U^{\mathrm{r}} = \max_{1 \leq j \leq J}(H_j - N_j), \qquad G_\pi^{\mathrm{r}} = U^{\mathrm{r}} - H_\tau + N_\tau. \tag{E.2}$$

Then the optimal stopping value is unchanged by removing date zero. Both M and N are integrable martingales, and

$$0 \leq G_\pi^{\mathrm{r}} \leq G_\pi \quad \text{almost surely}, \qquad \mathbb{E}[G_\pi^{\mathrm{r}} \mid \mathcal{D}] \geq \ell_\pi. \tag{E.3}$$

[5] Source: LSEG Workspace, accessed through the University of St Andrews; historical screening calculations are my own.

Proof. For any $\nu \in \mathcal{T}_{0:J}$, set $\nu' = \max(\nu, 1)$. The event $\{\nu' \le 0\}$ is empty and $\{\nu' \le j\} = \{\nu \le j\} \in \mathcal{F}_j$ for $j \ge 1$, so ν' is an admissible index in $\mathcal{T}_{1:J}$. Moreover,

$$H_{\nu'} - H_\nu = \mathbf{1}_{\{\nu=0\}} H_1 \ge 0. \tag{E.4}$$

Thus the restricted supremum is at least the full-calendar supremum. The reverse inequality follows from $\mathcal{T}_{1:J} \subseteq \mathcal{T}_{0:J}$, establishing equality.

Every X_j is integrable and adapted, with $\mathbb{E}[X_j \mid \mathcal{F}_{j-1}, \mathcal{D}] = 0$ by the defining conditional expectation. Consequently M is a martingale. The first increment of N is zero, and its subsequent increments are X_j, so N is also a martingale. For any $\nu \in \mathcal{T}_{1:J}$,

$$N_\nu = \sum_{r=2}^{J} X_r \mathbf{1}_{\{\nu \ge r\}}. \tag{E.5}$$

Here $\{\nu \ge r\}$ is $\mathcal{F}_{r-1}$-measurable because ν is a stopping index. The finite sum is integrable, and the tower property gives

$$\mathbb{E}[N_\nu \mid \mathcal{D}] = \sum_{r=2}^{J} \mathbb{E}\big[\mathbf{1}_{\{\nu \ge r\}} \mathbb{E}[X_r \mid \mathcal{F}_{r-1}, \mathcal{D}] \mid \mathcal{D}\big] = 0. \tag{E.6}$$

This verifies the finite-horizon optional-sampling step directly, including $J = 1$, when the sum is empty.

The maximum U^{r} is integrable since $|U^{\mathrm{r}}| \le \sum_{j=1}^{J} (|H_j| + |N_j|)$. For every retained stopping index, $U^{\mathrm{r}} \ge H_\nu - N_\nu$ pathwise. Taking expectations and the supremum therefore gives $\mathbb{E}[U^{\mathrm{r}} \mid \mathcal{D}] \ge V^*$. Applying the zero stopped expectation to τ yields

$$\mathbb{E}[G^{\mathrm{r}}_\pi \mid \mathcal{D}] = \mathbb{E}[U^{\mathrm{r}} \mid \mathcal{D}] - V_\pi \ge V^* - V_\pi. \tag{E.7}$$

Finally, $N_j = M_j - M_1$ for $j \ge 1$, whence

$$G^{\mathrm{r}}_\pi = \max_{1 \le j \le J} \{H_j - H_\tau - (M_j - M_\tau)\}. \tag{E.8}$$

The selected-date candidate is exactly zero. Every retained candidate also appears in the full-calendar maximum defining G_π, proving both pathwise inequalities in (19). □

F Clipped inference with numerical intervals

I use the following derivations to retain numerical uncertainty in a finite-sample monetary bound. For this inference, $\mathcal{D}$ contains the frozen fitting and design information and the audited pre-run exposed prefix; it excludes production gains and upper limits. From deterministic positive initial states, the model follows homogeneous independent GBM with $K > 0$, $r, q, \sigma \ge 0$, and finite calendar $0 < t_1 < \cdots < t_J = T$. The assessed $\tau \in \{1, \ldots, J\}$ is adapted to the own-path filtration containing $\mathcal{D}$ initially. The finite-horizon rewards and martingales are integrable. The derivations apply established norm and concentration inequalities to the assessed gain; they do not replace the model and sampling hypotheses.

For non-negative discounted max-call rewards and $B \ge 0$,

$$(G^0_\pi - B)^+ \le \sum_{j=1}^{J} (H_j - B)^+ \tag{F.1}$$

$$\le \sum_{j=1}^{J} \sum_{i=1}^{d} e^{-rt_j} (S^i_{t_j} - K - Be^{rt_j})^+. \tag{F.2}$$

The first inequality follows from $G_\pi^0 \le \max_j H_j$ and $(\max_j H_j - B)^+ \le \sum_j (H_j - B)^+$. The second follows by replacing the positive part of a coordinate maximum by the sum of coordinate positive parts. Taking conditional expectations gives a sum of marginal lognormal call moments. I evaluate discounts upwards and strike thresholds downwards, preserving the direction of this upper bound. The terminal reward remains in this envelope.

For the European gain, put $h_j = T - t_j$, $m_j = \max_i S_{t_j}^i$ and let $C_h(s)$ denote the time-h discounted European max-call value from state vector s. Choosing a coordinate attaining m_j and applying conditional Jensen gives

$$P_j \ge e^{-rt_j}(e^{-qh_j} m_j - e^{-rh_j} K)^+. \tag{F.3}$$

For $m_j \ge K$, using $(x)^+ \ge x$ implies

$$\begin{aligned} F_j^+ &\le \mathcal{L}_j^+ = (a_j m_j - \gamma_j)^+, \\ a_j &= e^{-rt_j}(1 - e^{-qh_j}), \qquad \gamma_j = e^{-rt_j} K (1 - e^{-rh_j}). \end{aligned} \tag{F.4}$$

For $m_j < K$, $H_j = 0$ and $F_j^+ = 0$, so the same upper envelope holds. With $\kappa_h = e^{-rh}\mathbb{E}[\max_i R_i]$ for the multiplicative terminal returns, monotonicity and the payoff Lipschitz bound give

$$C_h(s) \le C_h(K\mathbf{1}) + \kappa_h (m - K)^+, \qquad \kappa_h \le e^{-rh} + C_h(K\mathbf{1})/K. \tag{F.5}$$

Indeed, increasing every starting coordinate from K to $m \ge K$ increases the maximum payoff by at most $(m-K)\max_i R_i$. Also $\max_i R_i \le 1 + (\max_i R_i - 1)^+$, proving the second inequality. Hence

$$(-F_j)^+ \le \mathcal{L}_j^- = e^{-rt_j}\{\overline{C}_{h_j} + (\overline{\kappa}_{h_j} - 1)^+ (m_j - K)^+\}, \tag{F.6}$$

where bars are verified upper endpoints. Let $b_+, b_- \ge 0$. The case $m_j < K$ follows from monotonicity. Since $F_J = 0$, these envelopes give

$$\begin{aligned} G_\pi^{\mathrm{E}} &\le \max_j \mathcal{L}_j^+ + \max_j \mathcal{L}_j^-, \\ (G_\pi^{\mathrm{E}} - b_+ - b_-)^+ &\le \sum_j (\mathcal{L}_j^+ - b_+)^+ + \sum_j (\mathcal{L}_j^- - b_-)^+. \end{aligned} \tag{F.7}$$

Write $d_j = e^{-rt_j}\overline{C}_{h_j}$ and $\beta_j = e^{-rt_j}(\overline{\kappa}_{h_j} - 1)^+$. When $b_- \ge \max_j d_j$, define

$$k_j^+ = (\gamma_j + b_+)/a_j \quad (a_j > 0), \qquad k_j^- = K + (b_- - d_j)/\beta_j \quad (\beta_j > 0). \tag{F.8}$$

Then $(\mathcal{L}_j^+ - b_+)^+ = a_j (m_j - k_j^+)^+$ and $(\mathcal{L}_j^- - b_-)^+ = \beta_j (m_j - k_j^-)^+$. Replacing maximum-call payoffs by marginal sums gives the allowance

$$\begin{aligned} T_{b_+ + b_-}^{\mathrm{E}} = &\sum_{j: a_j > 0} \sum_i a_j \mathbb{E}[(S_{t_j}^i - k_j^+)^+ \mid \mathcal{D}] \\ &+ \sum_{j: \beta_j > 0} \sum_i \beta_j \mathbb{E}[(S_{t_j}^i - k_j^-)^+ \mid \mathcal{D}]. \end{aligned} \tag{F.9}$$

Zero slopes have exactly zero excess. Upper slopes and intercepts, lower γ_j, and downward-rounded thresholds preserve the upper direction of the numerical allowance. I use $b_+ = 20$ and $b_- = 30$. No fitted polynomial or hinge tail is substituted into this European envelope.

Finally suppose conditionally iid ideal clipped gains $Y_i \in [0, B]$ have valid, possibly data-dependent numerical enclosures $[l_i, u_i]$. Set $m_i = (l_i + u_i)/2$ and $r_i = (u_i - l_i)/2$. For the centring projection $P = I - \mathbf{1}\mathbf{1}^\top/n$, $\|P\|_2 = 1$. Write $Y = m + e$, where $|e_i| \le r_i$. The triangle inequality yields

$$\begin{aligned} s_Y = \frac{\|PY\|_2}{\sqrt{n-1}} \le \frac{\|Pm\|_2 + \|Pe\|_2}{\sqrt{n-1}} \le s_m + \sqrt{\frac{\sum_i r_i^2}{n-1}}, \\ \bar{Y} \le \frac{1}{n}\sum_i u_i. \end{aligned} \tag{F.10}$$

Applying Maurer and Pontil (2009, Theorem 4) to Y/B and adding the valid deterministic allowance $T_B \geq \mathbb{E}[(G-B)^+ \mid \mathcal{D}]$ proves (20), for $B > 0$ and $n \geq 2$. All transcendental operations and additions are rounded outwards. For $B = 0$, the upper construction is T_0 directly. For the amended inference, I additionally condition on the finite pre-run exposed prefix. The exclusion mask is determined solely by that earlier stream use. Under the maintained ideal independent-innovation premise, the remaining suffix retains its iid law; n counts only these eligible rows. Conditional on this information, frozen training, caps and assessors, each of the 32 events has failure probability at most δ. Their union has probability at most $32\delta = 0.05$, regardless of dependence between matched assessors or nested looks. Numerical refinement may depend on the observed path because the concentration event concerns the fixed ideal Y_i, while its enclosed mean and variance dominate it pointwise.

G Supported domains and boundary checks

The ordinary simulation accepts finite positive initial states, non-negative volatility and a symmetric positive-semidefinite correlation matrix, including singular cases. Exact zero-volatility transitions are deterministic. Regression policies specify ties, insufficient samples, rank deficiency and terminal exercise. The maximum conditional certificate has a narrower domain: independent assets, homogeneous strictly positive volatility and strictly positive hinge strikes and transition times. Unsupported correlation or a zero-volatility request is rejected instead of being passed through the product-CDF formula.

The gain calculation checks finite inputs, ordered intervals, calendar dimensions and valid stopping indices. A common initial martingale level cancels from every stopped difference. First, middle and last stopping indices, unsigned index arrays, signed coefficients, all-zero hinge branches and arithmetic overflow receive explicit checks. Interval decisions that remain ambiguous are reported; neither an arbitrary index array nor a passed finite suite proves universal admissibility or the absence of every implementation error.

H Probability foundations of the assessment

I give the conditional-expectation and measure-change arguments underlying regression and martingale assessment. The probability statements are for an independent evaluation path conditional on a fixed finite fitting record $\mathcal{D}$. All fitted-function moment statements are under this fixed-fit law. The results are standard; the derivation connects them to the objects calculated here (Williams, 1991; Shreve, 2004).

H.1 Conditional expectation and regression

Let $Y \in L^1(\mathbb{Q})$ and let $\mathcal{G}$ be a sub-σ-algebra of $\mathcal{F}$. The finite positive measures

$$\nu_\pm(A) = \mathbb{E}_\mathbb{Q}[Y^\pm \mathbf{1}_A], \qquad A \in \mathcal{G}, \tag{H.1}$$

are absolutely continuous with respect to $\mathbb{Q}|_\mathcal{G}$. The Radon–Nikodym theorem supplies non-negative $\mathcal{G}$-measurable densities $X_\pm$ with $\mathbb{E}_\mathbb{Q}[X_\pm] = \mathbb{E}_\mathbb{Q}[Y^\pm]$. Taking versions finite outside a null set, $X = X_+ - X_-$ is integrable and satisfies

$$\mathbb{E}_\mathbb{Q}[X\mathbf{1}_A] = \mathbb{E}_\mathbb{Q}[Y\mathbf{1}_A] \quad \text{for every } A \in \mathcal{G}. \tag{H.2}$$

This defines $X = \mathbb{E}_\mathbb{Q}[Y \mid \mathcal{G}]$ up to null sets. Uniqueness follows by applying the integral identity to the positive and negative sets of the difference of two candidate densities. The identity extends to bounded $\mathcal{G}$-measurable multipliers by simple-function approximation and dominated convergence (Williams, 1991, Chapter 9).

For $Y \in L^2(\mathbb{Q})$, set $X_n = X\mathbf{1}_{\{|X|\le n\}}$ and $a_n = \mathbb{E}_{\mathbb{Q}}[X^2\mathbf{1}_{\{|X|\le n\}}]$. Since X_n is a bounded permissible multiplier, Cauchy–Schwarz gives $a_n = \mathbb{E}_{\mathbb{Q}}[YX_n] \le \|Y\|_2\sqrt{a_n}$. Monotone convergence therefore yields $\|X\|_2 \le \|Y\|_2$, without first assuming conditional Jensen. For $Z \in L^2(\mathcal{G}, \mathbb{Q})$, bounded truncations Z_n converge in L^2 by dominated convergence. Cauchy–Schwarz extends $\mathbb{E}_{\mathbb{Q}}[(Y - X)Z_n] = 0$ to Z, giving

$$\mathbb{E}_{\mathbb{Q}}[(Y - Z)^2] = \mathbb{E}_{\mathbb{Q}}[(Y - X)^2] + \mathbb{E}_{\mathbb{Q}}[(X - Z)^2]. \tag{H.3}$$

Conditional expectation is consequently the unique mean-square predictor among $\mathcal{G}$-measurable candidates, up to null sets. The regression in the policy implementation restricts this projection to its finite basis. Its population approximation error and finite-sample fitting error remain distinct from the error of the resulting stopping decision.

H.2 The finite-calendar martingale argument

Frozen coefficients are measurable at the initial date. Evaluation innovations are independent of the fitting record, and exercise decisions use the current state and past information. If $M_0 = 0$ and M is an integrable martingale on this filtration, every stopping index $\tau \in \{0, \dots, J\}$ satisfies

$$\begin{aligned} M_\tau &= \sum_{j=1}^{J} \mathbf{1}_{\{\tau \ge j\}} \Delta M_j, \\ \mathbb{E}_{\mathbb{Q}}[M_\tau] &= \sum_{j=1}^{J} \mathbb{E}_{\mathbb{Q}}\big[\mathbf{1}_{\{\tau \ge j\}} \mathbb{E}_{\mathbb{Q}}[\Delta M_j \mid \mathcal{F}_{j-1}]\big] = 0. \end{aligned} \tag{H.4}$$

The indicator is $\mathcal{F}_{j-1}$-measurable because τ is a stopping index. Boundedness of the calendar and integrability of each increment justify the finite sum. No independence between τ and M is required. In the main text this zero expectation is conditional on $\mathcal{D}$; it is not a zero expectation conditional on the event $\{\tau = j\}$. Applying the same argument to any admissible index proves the dual upper inequality and the decomposition in (5). Appendix E gives the specialised calculation when the zero initial reward is removed.

H.3 Dividend convention and the pricing measure

For the nonsingular constant-coefficient case, let $\mathbf{B}^{\mathbb{P}}$ be a d-dimensional standard Brownian motion under a physical measure $\mathbb{P}$. I use its Brownian filtration under the fixed-fit evaluation law. Write the ex-dividend stock dynamics as

$$\frac{dS_t^i}{S_t^i} = (\mu_i - q_i)\, dt + \sum_k \Sigma_{ik}\, dB_t^{\mathbb{P},k}, \qquad \Gamma = \Sigma\Sigma^\top. \tag{H.5}$$

Here μ_i is the expected instantaneous total return, so the cash dividend $q_i S_t^i\, dt$ is subtracted from price drift. Assume Σ is invertible and put $\lambda = \Sigma^{-1}(\mu - r\mathbf{1})$. The density process

$$Z_t = \exp\{-\lambda^\top \mathbf{B}_t^{\mathbb{P}} - \tfrac{1}{2}\|\lambda\|^2 t\} \tag{H.6}$$

is strictly positive. The normal-increment moment formula gives $\mathbb{E}_{\mathbb{P}}[Z_t \mid \mathcal{F}_s] = Z_s$ and $\mathbb{E}_{\mathbb{P}}[Z_T] = 1$; equivalently Novikov's condition holds since $\exp(\|\lambda\|^2 T/2) < \infty$. Defining $d\mathbb{Q}/d\mathbb{P} = Z_T$ therefore gives equivalent measures on the finite horizon. Girsanov's theorem makes $\mathbf{B}_t^{\mathbb{Q}} = \mathbf{B}_t^{\mathbb{P}} + \lambda t$ Brownian under $\mathbb{Q}$, and substitution changes price drift to $r - q_i$ (Shreve, 2004). With $\Sigma = \operatorname{diag}(\sigma)L$ and $LL^\top = \rho$, this is exactly (1).

The defining integral identity also gives the conditional density formula

$$\mathbb{E}_{\mathbb{Q}}[Y \mid \mathcal{F}_s] = \frac{\mathbb{E}_{\mathbb{P}}[Z_T Y \mid \mathcal{F}_s]}{Z_s}, \qquad \mathbb{E}_{\mathbb{Q}}[|Y|] < \infty. \tag{H.7}$$

The right-hand side is integrable under $\mathbb{Q}$ because $\mathbb{E}_{\mathbb{P}}[|\mathbb{E}_{\mathbb{P}}[Z_T Y \mid \mathcal{F}_s]|] \le \mathbb{E}_{\mathbb{P}}[Z_T|Y|]$. For $A \in \mathcal{F}_s$, its $\mathbb{Q}$-integral equals the $\mathbb{P}$-integral weighted by Z_s, hence equals $\mathbb{E}_{\mathbb{P}}[Z_T Y \mathbf{1}_A]$. This verifies the defining conditional-expectation identity. I use the construction to explain the maintained pricing law; physical drifts are not estimated in the numerical study.

For singular Σ, the constant shift requires $\mu - r\mathbf{1} \in \mathrm{range}(\Sigma)$. An attainable drift permits a solution of $\Sigma\lambda = \mu - r\mathbf{1}$ and the same finite-horizon density argument, but invertibility and a general completeness claim are unavailable. A deterministic stock row requires $\mu_i = r$. The implementation also accepts singular or zero-volatility dynamics specified directly under a maintained $\mathbb{Q}$; that numerical branch does not by itself establish a physical-to-pricing measure construction.

H.4 Integrability and the PDE connection

For $S_0^i > 0$, the explicit solution under $\mathbb{Q}$ is

$$S_t^i = S_0^i \exp\{(r - q_i - \Gamma_{ii}/2)t + \Sigma_{i\cdot}\mathbf{B}_t^{\mathbb{Q}}\}. \tag{H.8}$$

Itô's formula verifies this solution. Linear drift and diffusion coefficients are globally Lipschitz with linear growth, giving pathwise uniqueness. Applying Itô to $\log S^i$ recovers the transition used in simulation. For any finite real vector p,

$$\mathbb{E}_{\mathbb{Q}}\left[\prod_i (S_t^i)^{p_i}\right] = \prod_i (S_0^i)^{p_i} \exp\left\{t\sum_i p_i(r - q_i - \tfrac{1}{2}\Gamma_{ii}) + \tfrac{t}{2}p^\top \Gamma p\right\}. \tag{H.9}$$

I obtain this identity from the Gaussian exponential moment. Under independence it reduces to the implemented polynomial centres and verifies square integrability of finite-degree polynomials, their sorted-coordinate versions and the finite linear-growth hinge fits at a fixed finite fit. Max-call rewards have the same property since $(\max_i S^i - K)^+ \le \sum_i S^i$ for $K \ge 0$. The finite number of rights then makes the stopped rewards and the dual maximum integrable.

Discounted price alone is not the dividend-paying stock's martingale. Its discounted cumulative gains satisfy

$$d\left(e^{-rt}S_t^i + \int_0^t e^{-ru} q_i S_u^i \, du\right) = e^{-rt} S_t^i \sum_k \Sigma_{ik} \, dB_t^{\mathbb{Q},k}. \tag{H.10}$$

The second moments in (H.9) make this a true martingale. For a smooth continuation value v, a self-financing portfolio with $\Delta_i = v_{s_i}$ has gains

$$d\Pi_t = \sum_i \Delta_i (dS_t^i + q_i S_t^i \, dt) + r\left(v - \sum_i \Delta_i S_t^i\right) dt. \tag{H.11}$$

Matching the Brownian coefficients in Itô's formula and cancelling the physical stock drift gives

$$v_t + \tfrac{1}{2}\sum_{i,k} \Gamma_{ik} s_i s_k v_{s_i s_k} + \sum_i (r - q_i) s_i v_{s_i} - rv = 0. \tag{H.12}$$

On an interval $[a, b]$ without exercise, the condition

$$\mathbb{E}_{\mathbb{Q}} \int_a^b e^{-2rt} \|\nabla v(t, S_t)^\top \mathrm{diag}(S_t)\Sigma\|^2 dt < \infty \tag{H.13}$$

turns the discounted value's local martingale into a true martingale and yields the conditional discounted-payoff representation. In the nonsingular case at positive time to expiry, Gaussian smoothing gives a classical continuation value. The European max-call has coordinate deltas bounded by $e^{-q_i(T-t)}$, so the lognormal second moments verify the condition. Singular covariance can retain payoff kinks and cannot inherit that smoothness argument. A Bermudan continuation uses this equation between rights and the maximum of continuation and immediate reward at a right. The one-asset case reduces to the Black–Scholes PDE used by the reference solver (Black and Scholes, 1973; Shreve, 2004).

I Regression and the numerical pricing reference

I.1 Three distinct least-squares objects

Equation (H.3) separates the conditional mean $C = \mathbb{E}[Y \mid \mathcal{G}]$ from its prediction residual. I distinguish this population target, its finite-basis approximation and its sample fit. For square-integrable basis functions $\phi = (\phi_0, \dots, \phi_{p-1})^\top$, minimisation over $\phi^\top\beta$ instead gives $\mathbb{E}[\phi\phi^\top]\beta = \mathbb{E}[\phi Y]$. The finite-dimensional subspace is closed, so its projected function is unique, even when linear dependence makes its coefficients nonunique. A basis need not contain C.

For observed responses y, design matrix X and positive weights $W = \operatorname{diag}(w_i)$, the empirical criterion is $(X\beta - y)^\top W(X\beta - y)$. Differentiation gives $X^\top WX\beta = X^\top Wy$. If $W^{1/2}X = U\operatorname{diag}(\varsigma_k)V^\top$, its pseudoinverse supplies minimum-norm least-squares coefficients. The reported regressions use $W = I$. At full column rank, $\kappa_2(X^\top WX) = \kappa_2(W^{1/2}X)^2$. Forming normal equations therefore aggravates conditioning; the policy implementations use the least-squares routine's rank and singular-value diagnostics rather than solving a Gram system directly.

The one-asset implementation centres and scales current spots, reduces polynomial degree until rank is full and the condition number is at most 10^{10}, and continues when fewer than two observations are available. The multiasset baseline uses descending sorted coordinates $S^i/K - 1$, retains its requested degree only at full rank, and has no condition-number rejection threshold. Its nonconstant fit needs at least $\max(64, p + 1)$ eligible observations under the default setting. When immediate reward is positive at time zero, it uses mean training continuation instead of fitting a constant-state design. Sorting preserves the exchangeable benchmark's symmetry; it can discard relevant asset identity under heterogeneity. Both policies clip negative continuation predictions at zero and continue on exercise ties. These are distinct implementations of the established backward regression principle of Longstaff and Schwartz (2001). Their response is the reward selected by the already fitted later policy, rather than the exact future Snell value. Regression accuracy, later-policy error and exercise quality consequently remain separate.

I.2 Backward PDE propagation and its error sources

In one asset, (H.12) has generator $\mathcal{L}v = (r - q)sv_s + \sigma^2 s^2 v_{ss}/2$. Between exercise rights, reverse time gives $u_\tau = Lu$, with $L = \mathcal{L} - rI$. At a right, I replace continuation by $\max(g, u)$. This separates diffusion propagation from the discrete exercise decision.

On $s_i = i\Delta s$, the interior matrix L_h has lower, upper and diagonal entries

$$\ell_i = \tfrac{1}{2}\sigma^2 i^2 - \tfrac{1}{2}(r - q)i, \quad b_i = \tfrac{1}{2}\sigma^2 i^2 + \tfrac{1}{2}(r - q)i, \quad d_i = -\ell_i - b_i - r. \tag{I.1}$$

Taylor expansion gives second-order spatial consistency where the value has bounded derivatives through order four. If either off-diagonal entry is negative, the code replaces them by $\ell_i = \sigma^2 i^2/2 + \max(-(r - q)i, 0)$ and $b_i = \sigma^2 i^2/2 + \max((r - q)i, 0)$. This preserves nonnegative transition coefficients but introduces first-order upwind drift error at those nodes.

A backward step of length h solves the tridiagonal system

$$(I - \theta h L_h)u^{\text{new}} = (I + (1 - \theta)hL_h)u^{\text{old}} + b^{\text{boundary}}, \qquad \tfrac{1}{2} \le \theta \le 1. \tag{I.2}$$

For smooth solutions, Crank–Nicolson has second-order temporal consistency; implicit Euler has first order. These orders do not apply automatically at the payoff kink or immediately after an exercise projection. Two implicit half steps replace the first full step after expiry and each retained exercise projection. I use the established damping construction discussed by Giles and Carter (2006), without inferring a global convergence order solely from that choice.

For $r \ge 0$, nonnegative ℓ_i, b_i make $I - \theta h L_h$ strictly diagonally dominant with nonpositive off-diagonal entries and a nonnegative inverse. Indeed, a negative minimum in a vector z would make the corresponding component of $(I - \theta h L_h)z$ negative; thus a nonnegative right-hand side cannot have such a solution. Implicit Euler is therefore positivity preserving for nonnegative rewards and

boundary values. For $\theta < 1$, entrywise nonnegativity of the explicit matrix supplies a sufficient positivity condition. A sufficient step restriction is $(1-\theta)h(\ell_i+b_i+r) \leq 1$ at every interior node. Under this restriction the homogeneous step has maximum-norm bound $\{1-(1-\theta)hr\}/(1+\theta hr) \leq 1$: the explicit matrix is nonnegative with row sums at most the numerator, while diagonal dominance bounds the implicit inverse by the reciprocal denominator. This compares runs with identical prescribed boundaries. Changed boundaries add $(I-\theta hL_h)^{-1}\Delta b^{\text{boundary}}$ to the error recursion; both old and new boundary values enter that forcing. These matrix bounds concern exact arithmetic; the linear solve also incurs rounding error. Crank–Nicolson's scalar stability property alone does not establish positivity. Negative rates, which the input model permits, require a separate growth and time-matrix assessment; the preceding contraction argument concerns $r \geq 0$. The exercise projection is nonexpansive in the maximum norm because $|\max(g,x)-\max(g,y)| \leq |x-y|$ for an unchanged payoff.

At zero stock, the absorbing GBM boundary permits exact optimisation over the remaining exercise dates. At the upper truncated boundary, the code sets a put to zero or optimises the call's discounted linear asymptote. These upper boundaries neglect return across the strike. I therefore assess domain expansion at unchanged spatial spacing separately from mesh refinement. Linear interpolation at the valuation spot and finite-domain error also enter the reference value. Agreement of refined grids is evidence of numerical resolution, rather than a certified diffusion price enclosure.

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